Equivalence Bridges

from the 130-Manuscript Corpus

Explicit independence, arithmetic obstructions,
and a positivity boundary

Prepared for Keio

Abstract. This note develops seven connections from selected arguments in the Verifiable Mathematical Reasoning Corpus. Its 28 numbered theorems and corollaries include a complete algebraic-relation criterion for logarithm-cancelled Bessel values; density-one independence under rational height; exact prime-power multiplicity and a unique height-eight minimizer; a quadratic weighted-binary denominator obstruction; a number-field extension of the translated reciprocal-tail method; geometric characterizations of universal integrability; and an exact contact description of a Le Roy complete-monotonicity boundary.

Three representative conclusions are

$$\operatorname{trdeg}_{\mathcal{F}} \mathcal{F}(C\kappa) = \operatorname{rank}_{\mathbb{Q}} C, \qquad \lambda^d \le d[K:\mathbb{Q}] + 1,$$

and the certified strict lower bound

$$\beta_c(1/3, 2) > 0.428292245841.$$

Each statement is given with its hypotheses and proof. Exact finite checks and interval certificates accompany the analytic arguments. The count includes supporting corollaries and useful classical equivalences; it is not a claim to 28 independent discoveries or to 28 newly solved conjectures. No historical priority or external peer-review certification is asserted.

Corpus snapshot: `c30d40119134a59eda6fa5cc6331f27c8e67d034`.

All 130 manuscript files matched the snapshot's Git blob hashes.

Mathematical review was targeted at the inputs used in this note.

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1 Sources, results, and proof boundaries

1.1 Verified source scope

The source repository is the Verifiable Mathematical Reasoning Corpus. Its immutable snapshot contains 130 numbered `.tex` manuscripts. On 8 November 2026, all 130 local file contents were compared with the current repository tree using Git's blob hash rule and matched. This checks source identity and coverage of the inventory; it does not verify every argument in all 130 manuscripts.

The principal inputs inspected for this synthesis are the following. The bracketed numbers throughout the note refer to corpus manuscript numbers, while ordinary bibliography citations refer to external works.

Source	Subject	Role here
[044]	Euler–Gompertz and Bessel values	Joint E-function independence and affine logarithmic rank
[056]	Square moduli and vanishing Erdős sums	Congruence-constrained periodic coefficients
[080]	Exponent window for Erdős 265	Positive multilinear tails
[086]	Three nearby primes in Erdős 700	Lucas digit restrictions
[088]	Integer-tail analytic attack	Perturbation classes and the Pólya–Carlson bridge
[112]	Period obstructions for weighted binary expansions	Multiplicative orders and valuations
[113], [114]	Polynomial inverses and integration stability	Polynomial primitive towers and geometric obstructions
[116]	Le Roy positivity beyond infinite divisibility	Inverse Mellin density and the strict positivity gap

For Bessel values, Section 2.1 reconstructs the positive-scale joint-independence argument and states separately the stronger six-coordinate input from [044]. The linear-algebra and height deductions are then proved here. The discrete, norm, geometric, and positivity arguments below include the steps on which their new conclusions depend. Classical specialization, de Rham, moment, and natural-boundary theorems remain explicitly identified mathematical inputs, rather than being counted as new proofs.

1.2 Seven connections and their exact outputs

Theme	Count	Main output
Bessel independence	5	All polynomial relations among cancelled coordinates are exactly the row relations of a rational matrix; explicit infinite independent family and height bounds.
Periodic harmonic sums	5	Exact attainable zero multiplicities, congruence iff tests, and a unique height-optimal period-64 example.
Quadratic binary coding	5	Prime-cube denominator obstruction, finite period bounds, and recurrence/natural-boundary equivalences.
Binomial gcd values	1	Exact four-value distribution, including all multiplicities, in the nearby-prime family.
Translated reciprocal sums	2	Common number-field degree lower bounds, uniformly across the perturbation class of Erdős 263.
Algebraic integration	3	Obstruction dimension $2g + s - 1$ and universal algebraic/elementary integrability criteria.
Le Roy positivity	7	Largest-contact equivalence, strict finite Hankel hierarchy, certified numerical lower bound, and convex-order comparison.

Reading the count. There are 28 theorem/corollary statements, in addition to auxiliary lemmas. For example, the six-way equivalence for binary sequences is a classical analytic consequence used to

transfer the new quadratic denominator obstruction to recurrence theory. The universal integration criteria likewise make classical geometric machinery explicit. "Additional" means an additional conclusion relative to the selected corpus passages, not a verified worldwide novelty claim.

Connections to open questions. The number-field result covers the entire relative-perturbation class asked about in Erdős 263, while leaving its singled-out rationality question unresolved. The Le Roy analysis characterizes the unknown true boundary without giving its closed form. The Bessel results give explicit independent combinations while preserving the distinction between combinations and individual raw second-kind values. The final section identifies the precise remaining steps, rather than treating an equivalence itself as a solution of its open endpoint.

Literature controls. The CM–moment connection is already in Simon [3]. E-function specialization is classical [2]. The unperturbed Mahler/Fermat reciprocal values have classical transcendence results [8, 9]; they are not the new perturbative conclusion here. Recent cumulant lifting results [10] also make it inappropriate to count a generic Bell/Hankel reformulation as a new conjecture resolution.

2 An exact algebraic-relation criterion for Bessel circuits

Put

$$k = \overline{\mathbb{Q}}, \qquad K_{\exp} = k(e^a : a \in k), \qquad \kappa(t) = \frac{K_0(t)}{I_0(t)} \quad (t > 0),$$

and let

$$\mathcal{F} = K_{\exp}\big(I_0(t), I_1(t) : t \in k \cap \mathbb{R}_{>0}\big).$$

All assertions over these countably generated fields have their usual finite-subset meaning. The harmonic Bessel coordinate is

$$\beta(t) = \frac{\displaystyle\sum_{n\geq 0} H_n (t/2)^{2n}/(n!)^2}{I_0(t)}, \qquad H_0 = 0.$$

The connection formula is

$$\beta(t) = \gamma + \log(t/2) + \kappa(t). \tag{2.1}$$

The positive-argument part of corpus [044, Theorem `cf:joint`] says that the entire family $\{\beta(t) : t \in k \cap \mathbb{R}_{>0}\}$ is algebraically independent over $\mathcal{F}$. We use this precise statement, not an inference from separate one-point transcendence results. A verification of the required functional argument is given below.

Theorem 2.1 (Complete relation ideal for cancelled coordinates)**.** *Let $t_1, \dots, t_m$ be distinct positive algebraic numbers. Let $C = (c_{ij}) \in \mathbb{Q}^{s\times m}$ satisfy, for every row,*

$$\sum_{j=1}^{m} c_{ij} = 0, \qquad \prod_{j=1}^{m} t_j^{c_{ij}} = 1.$$

Define

$$L_i = \sum_{j=1}^{m} c_{ij}\kappa(t_j), \qquad 1 \leq i \leq s.$$

Then

$$\operatorname{trdeg}_{\mathcal{F}} \mathcal{F}(L_1, \dots, L_s) = \operatorname{rank}_{\mathbb{Q}} C. \tag{2.2}$$

More precisely, the kernel of evaluation $\mathcal{F}[X_1, \dots, X_s] \to \mathbb{C}$, $X_i \mapsto L_i$, is the ideal generated by

$$\sum_i u_i X_i, \qquad u \in \mathbb{Q}^s, \quad u^{\mathsf{T}} C = 0.$$

In particular, the L_i are algebraically independent over $\mathcal{F}$ if and only if the rows of C are linearly independent over $\mathbb{Q}$.

Proof. For positive arguments the multiplicative equation is equivalent to $\sum_j c_{ij} \log t_j = 0$. Equation (2.1) therefore gives $L_i = \sum_j c_{ij}\beta(t_j)$. The $\beta(t_j)$ are algebraically independent over $\mathcal{F}$, so they can be treated as indeterminates. Let $r = \operatorname{rank} C$. Rational row operations and an invertible rational change of coordinates on the indeterminates turn the map $X \mapsto C\beta$ into

$$(X_1, \dots, X_s) \longmapsto (Y_1, \dots, Y_r, 0, \dots, 0).$$

Its kernel is generated by $X_{r+1}, \dots, X_s$. Undoing the changes of coordinates proves both assertions. □

This is a full algebraic-relation classification for the stated class of expressions. It does not assert that an arbitrary combination of second-kind Bessel values satisfies the two cancellation conditions. For rational arguments the conditions are decidable by exact prime factorization: use the matrix whose columns are $(1, (v_p(t_j))_p)$.

Corollary 2.2 (A canonical explicit independent family)**.** *For every positive rational r other than* 1 *or a prime, put*

$$D_r = \kappa(r) - \kappa(1) - \sum_p v_p(r)(\kappa(p) - \kappa(1)). \tag{2.3}$$

The sum is finite. The entire family

$$\{D_r : r \in \mathbb{Q}_{>0} \setminus (\{1\} \cup \{primes\})\}$$

is algebraically independent over $\mathcal{F}$. In particular every member is transcendental. Any nonconstant rational function, with coefficients in $\mathcal{F}$, of finitely many distinct members is transcendental over $\mathcal{F}$.

Proof. Unique prime factorization gives $\log r = \sum_p v_p(r) \log p$, and the coefficients in (2.3) sum to zero. In any finite list of such rows, the column indexed by its non-anchor argument r has coefficient one in that row and zero in the others. The rows are therefore independent. Apply Theorem 2.1. A rational function field is relatively algebraically closed over its coefficient field, which proves the last assertion. □

For example, the following four explicitly specified numbers are jointly algebraically independent, even over $\mathcal{F}$:

$$\begin{aligned} D_4 &= \kappa(4) + \kappa(1) - 2\kappa(2), \\ D_6 &= \kappa(6) + \kappa(1) - \kappa(2) - \kappa(3), \\ D_9 &= \kappa(9) + \kappa(1) - 2\kappa(3), \\ D_{1/2} &= \kappa(1/2) + \kappa(2) - 2\kappa(1). \end{aligned}$$

For illustration,

$$(D_4, D_6, D_9) \approx (0.2336077763,\ 0.2754838101,\ 0.3183102998).$$

These numerical values play no role in the proof.

2.1 Verification of the independence input used here

Here is a short reconstruction of the positive-scale input of [044]. It explains why the argument requires joint independence and why rationally dependent scales such as $1, 2, 4$ cause no difficulty. For a fixed nonzero algebraic scale t, write

$$f_t(z) = I_0(tz), \qquad y_t(z) = \sum_{n \geq 0} \frac{H_n (tz/2)^{2n}}{(n!)^2}.$$

Then

$$f_t'' + z^{-1} f_t' - t^2 f_t = 0, \qquad y_t'' + z^{-1} y_t' - t^2 y_t = 2 f_t'/z,$$

and $g_t = y_t - f_t \log z$ is a second homogeneous solution, with Wronskian $f_t g_t' - f_t' g_t = -1/z$.

The homogeneous equation has differential Galois group SL_2. One can check the usual Bessel input directly: its logarithmic local solution gives nontrivial unipotent monodromy, and it is irreducible over $\mathbb{C}(z)$. Indeed a rational logarithmic derivative r would satisfy $r' + r^2 + r/z - t^2 = 0$. At every nonzero finite pole its residue would be one, while it has no pole at zero. At infinity it would have expansion $r = \pm t - 1/(2z) + O(z^{-2})$, contradicting the nonnegative integer sum of its finite residues. The classification of algebraic subgroups of SL_2, together with the unipotent element, then excludes every proper irreducible subgroup.

For scales distinct modulo sign the joint group is the product SL_2^m. If a subdirect product were smaller, two of its adjoint modules would become isomorphic after a finite algebraic base change. Their exponential parts at infinity are respectively $0, 2t_i, -2t_i$ and $0, 2t_j, -2t_j$. An invertible formal intertwiner forces equality of these sets, and hence $t_i = \pm t_j$, a contradiction. This is the argument in [044, Lemma `cf:product`].

Adjoin any finite collection of exponential functions with algebraic exponents and $\log z$. Their Picard–Vessiot group is solvable, whereas SL_2^m has no nontrivial solvable quotient. The groups therefore form a direct product. Since the full Bessel solution field is recovered from f_t, f_t', y_t after adjoining $\log z$, dimension counting gives functional algebraic independence of these $3m$ coordinates over the exponential field. All f_t, f_t', y_t, y_t' are E-functions in a first-order system whose only finite singularity is zero. Ordinary-point Siegel–Shidlovskii specialization at $z = 1$ applies [2]. The identity

$$f_t' y_t - f_t y_t' = (1 - f_t^2)/z$$

recovers $y_t'(1)$ from the other three values, since $I_0(t) > 0$ for positive t. Lindemann–Weierstrass gives the expected dimension of the exponential value field. Thus the $3m$ values are jointly independent over each finite exponential coefficient field. Passing to I_0, I_1, β is birational. Every proposed relation over $\mathcal{F}$ uses only finitely many additional scales and exponentials, so the same argument proves the claimed infinite-base independence. The specialization input is classical; the scaled product argument is explicitly checked here and in [044].

3 From integer arguments to rational height and number fields

For a finite set $T = \{t_1, \dots, t_m\}$ of distinct positive algebraic numbers, define

$$W_T = \{c \in \mathbb{Q}^m : \sum_j c_j = 0,\ \sum_j c_j \log t_j = 0\}, \qquad d_T = m - \dim_{\mathbb{Q}} W_T.$$

Theorem 2.1 gives $m - d_T$ independent linear combinations of $\kappa(t_j)$ over $\mathcal{F}$. Consequently

$$\operatorname{trdeg}_{K_{\exp}} K_{\exp}(K_0(t) : t \in T) \geq |T| - d_T. \tag{3.1}$$

This is the affine-rank theorem in [044, Theorem `kd:rank`], now with an exact relation-ideal formulation for its cancelled outputs.

The stronger six-coordinate input of [044, Theorem `jden:rank`] concerns

$$\begin{aligned} Z_\lambda &= \big(E_1^{\mathrm{tail}}(\lambda), K_0(2\lambda), K_1(2\lambda), \pi Y_0(2\lambda), \pi Y_1(2\lambda), \mathcal{L}(\lambda)\big), \\ E_1^{\mathrm{tail}}(\lambda) &= \int_\lambda^\infty e^{-u}\, du/u, \\ \mathcal{L}(\lambda) &= \sum_{n \geq 1} \frac{(-1)^{n-1} \lambda^n}{n (n!)^2} - \gamma - \log \lambda. \end{aligned}$$

It states

$$\operatorname{trdeg}_{K_{\exp}} K_{\exp}(Z_\lambda : \lambda \in T) \geq 6|T| - d_T. \tag{3.2}$$

For clarity, this uses more than the positive-scale Bessel input: it also uses the ordinary-Bessel scales, the entire exponential integral, and the primitive nonexactness lemma `msprim:nonexact` in [044]. The latter reduces to the impossibility of a rational solution of

$$-b'' + b'/z - b/z^2 + t^2 b = 1/z.$$

Nonzero finite poles are impossible. At zero a nonpositive lowest Laurent exponent produces an uncancellable pole of order at least two; a positive lowest exponent produces no z^{-1} term. This verifies the critical rational obstruction in that source. The algebraic recovery in [044, `jden:rank`] uses eight independent E-function coordinates, two additional first-kind coordinates, and the common logarithmic family $\gamma + \log\lambda$, of rank at most d_T. Thus $8|T| \leq \operatorname{trdeg}(Z_T) + 2|T| + d_T$, which is (3.2).

Theorem 3.1 (Rational-height density and simultaneous independence)**.** *Let*

$$\mathcal{R}_N = \{a/b : 1 \leq a, b \leq N,\ \gcd(a,b) = 1\}, \qquad M_N = |\mathcal{R}_N| = 2\sum_{n=1}^{N} \varphi(n) - 1.$$

Then

$$\operatorname{trdeg}_{K_{\exp}} K_{\exp}(K_0(r) : r \in \mathcal{R}_N) \geq M_N - \pi(N) - 1, \tag{3.3}$$

$$\operatorname{trdeg}_{K_{\exp}} K_{\exp}(Z_{r/2} : r \in \mathcal{R}_N) \geq 6M_N - \pi(N) - 1. \tag{3.4}$$

There is a single subset $\mathcal{S} \subset \mathbb{Q}_{>0}$ with

$$|\mathcal{S} \cap \mathcal{R}_N| \geq M_N - \pi(N) - 1 \qquad (N \geq 1)$$

such that all six coordinates of $Z_{r/2}$, as r ranges over $\mathcal{S}$, are jointly algebraically independent over $K_{\exp}$. At most $\pi(N)+1$ arguments $r \in \mathcal{R}_N$ can have any one of

$$E_1^{\mathrm{tail}}(r/2), \quad \mathcal{L}(r/2), \quad K_j(r), \quad \pi Y_j(r) \quad (j \geq 0)$$

algebraic over $K_{\exp}$.

Proof. The logarithms of the rationals in $\mathcal{R}_N$ span exactly the rational space generated by $\log p$ for primes $p \leq N$. The augmented exponent vectors of 1 and of these primes are independent. Hence $d_{\mathcal{R}_N} = \pi(N) + 1$. Common multiplication by $1/2$ preserves affine logarithmic rank, so (3.1) and (3.2) give the two bounds. The counting formula for M_N follows by counting the coprime pairs with $a \leq b$, and then reflecting across the diagonal; only $(1,1)$ is on the diagonal.

Enumerate positive rationals by nondecreasing $\max(a,b)$, and process the six coordinates in each block in a fixed order. Keep a coordinate exactly when it is transcendental over the field generated by the preceding kept coordinates and $K_{\exp}$. At the end of $\mathcal{R}_N$, at most $\pi(N)+1$ coordinates have been rejected by (3.4). Delete every block containing a rejected coordinate. At most the same number of blocks is deleted, and all remaining coordinates are jointly independent.

For the final statement, let T be the set of bad arguments. The standard integer-order recurrences express every $K_j(r)$, respectively $\pi Y_j(r)$, as a nonzero algebraic linear form in the corresponding orders zero and one. Thus each bad block has a polynomial relation and transcendence degree at most five. If $m = |T|$, its total rank is at most $5m$, whereas (3.2) gives at least $6m - d_T$. Therefore $m \leq d_T \leq \pi(N) + 1$. □

The independent set is existential; this proof does not provide a membership test or identify an individual algebraic or transcendental raw second-kind value. It does, however, give a much sparser exception proportion under rational height:

$$M_N = \frac{6}{\pi^2} N^2 + O(N \log N), \qquad \frac{\pi(N)+1}{M_N} = O\!\left(\frac{1}{N \log N}\right).$$

For example,

N	M_N	block deletion bound	retained blocks	six-coordinate rank bound
10	63	5	58	373
100	6087	26	6061	36496
1000	608383	169	608214	3650129

The numerical table uses only exact totient and prime counts.

Corollary 3.2 (Stability under finitely many additional constants)**.** *Let $\mathcal{B} \supset K_{\exp}$ be a subfield of $\mathbb{C}$ with finite $d = \operatorname{trdeg}_{K_{\exp}} \mathcal{B}$. Every bound in Theorem 3.1 remains valid over $\mathcal{B}$ after adding d to its deletion or exception budget, or subtracting d from its rank bound. In particular, adjoining any fixed finite list of unresolved constants preserves the density-one conclusion.*

Proof. For any finite tuple x,

$$\operatorname{trdeg}_{\mathcal{B}} \mathcal{B}(x) \geq \operatorname{trdeg}_{K_{\exp}} K_{\exp}(x) - d.$$

Apply this to the rank inequalities, then repeat the same greedy argument and bad-block count. A list of d_0 constants gives $d \leq d_0$, irrespective of their presently unknown arithmetic natures. □

Theorem 3.3 (Density one in a fixed real-embedded number field)**.** *Let K be a number field of degree d_K with a fixed real embedding σ_0, and let $u_K = r_1 + r_2 - 1$ be its unit rank. Identify its elements with their images under σ_0, and define*

$$\mathcal{A}_X = \{\alpha \in \mathcal{O}_K : \sigma_0(\alpha) > 0,\ |\sigma(\alpha)| \leq X \text{ for every embedding } \sigma : K \hookrightarrow \mathbb{C}\}.$$

Put $M_K(X) = |\mathcal{A}_X|$ and

$$B_K(X) = d_K \pi(X^{d_K}) + u_K + 1.$$

Then

$$\operatorname{trdeg}_{K_{\exp}} K_{\exp}(K_0(\alpha) : \alpha \in \mathcal{A}_X) \geq M_K(X) - B_K(X),$$
$$\operatorname{trdeg}_{K_{\exp}} K_{\exp}(Z_{\alpha/2} : \alpha \in \mathcal{A}_X) \geq 6M_K(X) - B_K(X).$$

There is a single set of retained arguments having at least $M_K(X) - B_K(X)$ members in every $\mathcal{A}_X$, on which all six coordinates are jointly independent. The retained set has relative density one in these expanding regions. The same conclusion survives a fixed finite-transcendence-degree coefficient extension.

Proof. For $\alpha \in \mathcal{A}_X$, $0 < |N_{K/\mathbb{Q}}(\alpha)| \leq X^{d_K}$. Every prime ideal dividing (α) lies over a rational prime at most X^{d_K}. There are at most $d_K \pi(X^{d_K})$ such prime ideals. The valuation map from the multiplicative group supported on these primes to the free abelian group on the primes has kernel $\mathcal{O}_K^\times$. Consequently its rank is at most $d_K \pi(X^{d_K}) + u_K$. No assumption that the prime ideals are principal is needed. Thus $d_{\mathcal{A}_X} \leq B_K(X)$, and the rank and greedy arguments apply exactly as before.

The Minkowski embedding of $\mathcal{O}_K$ is a lattice of covolume $2^{-r_2}\sqrt{|\Delta_K|}$. The region consists of a half-box in the real coordinates and disks in the complex coordinates. Elementary lattice counting gives

$$M_K(X) = \frac{2^{r_1+r_2-1}\pi^{r_2}}{\sqrt{|\Delta_K|}} X^{d_K} + O_K(X^{d_K-1}).$$

On the other hand $B_K(X) = O_K(X^{d_K}/\log X)$. Their ratio tends to zero. Finally use Corollary 3.2. □

4 Prime-power coefficient congruences and high-order harmonic zeros

The congruence classification of corpus [056] concerns the existence of rational Okada vectors. The following statements refine the prime-power construction by measuring the order of vanishing of the entire periodic Dirichlet series at $s = 1$. Their scope is explicit: the first theorem classifies a separated divisor-layer family, and the second classifies functions constant on each gcd layer. Neither is a classification of binary sign functions at arbitrary square moduli.

Let ℓ and p be primes with $p \equiv 1 \pmod{\ell}$. Define the p-periodic integer function

$$u(0) = 0, \qquad u(1) = 2 - p, \qquad u(a) = 1 \quad (2 \le a \le p - 1).$$

Its period sum is zero and all its nonzero-residue values are one modulo ℓ. Furthermore

$$L(1, u) < 0. \tag{4.1}$$

Indeed, each length-p block is the sum of $p - 2$ positive reciprocals with denominators larger than the denominator of the negative term of magnitude $p - 2$. Thus each block is strictly negative, and their sum converges.

Theorem 4.1 (Exact maximal zero order in separated prime-power layers)**.** *Let $e \ge 1$. For rational $c_0, \dots, c_{e-1} \in 1 + \ell\mathbb{Z}_{(\ell)}$, define the p^e-periodic function*

$$f(n) = \begin{cases} c_t u(n/p^t), & v_p(n) = t < e, \\ 0, & p^e \mid n. \end{cases}$$

Then its period sum is zero, and

$$L(s, f) = C(p^{-s})L(s, u), \qquad C(T) = \sum_{t=0}^{e-1} c_t T^t.$$

The maximum possible order of vanishing at $s = 1$, as the c_t vary, is exactly

$$\ell^{v_\ell(e)} - 1. \tag{4.2}$$

For every integer $r \ge 1$, the following are equivalent:

1. *At least one such f satisfies $L^{(j)}(1, f) = 0$ for $0 \le j < r$.*
2. *$\ell^{v_\ell(e)} \ge r + 1$.*
3. *$\ell^{\lceil \log_\ell(r+1) \rceil}$ divides e.*

The maximal order is attained with integer coefficients and exact fundamental period p^e.

Proof. The Dirichlet-series identity follows by partitioning the summation according to $v_p(n)$. Each divisor layer has zero mean because u has zero mean. By (4.1), the zero order is exactly the root multiplicity of $C(T)$ at $T = 1/p$.

Modulo ℓ, one has $\overline{C}(T) = S_e(T) = 1 + T + \cdots + T^{e-1}$, and $1/p$ reduces to one. Write $e = Dm$, where $D = \ell^{v_\ell(e)}$ and $\ell \nmid m$. In characteristic ℓ,

$$S_e(T) = \frac{(T^m - 1)^D}{T - 1}$$

has a root of exact multiplicity $D - 1$ at one. Since p is an ℓ-adic unit, any factor $(T - 1/p)^r$ of C over the rationals can be divided out over $\mathbb{Z}_{(\ell)}$; reduction therefore gives $r \le D - 1$.

Equality is attained by

$$C(T) = (1 - pT)^{D-1} \sum_{j=0}^{m-1} (pT)^{Dj}. \tag{4.3}$$

Its reduction is $(1 + T + \cdots + T^{D-1}) \sum_{j=0}^{m-1} T^{Dj} = S_e(T)$, so every coefficient is one modulo ℓ. The second factor has value $m \neq 0$ at $T = 1/p$. This proves the exact order and the equivalences. No off-zero coefficient of f vanishes, since it is one modulo ℓ. As p^e is a period and its zero set is exactly the multiples of p^e, its fundamental period is p^e. □

Corollary 4.2 (Three logarithmic harmonic cancellations)**.** *There is an odd-integer-valued periodic function of exact period* 81*, zero exactly at multiples of* 81*, such that*

$$\sum_{n\geq 1} \frac{f(n)}{n} = \sum_{n\geq 1} \frac{f(n)\log n}{n} = \sum_{n\geq 1} \frac{f(n)(\log n)^2}{n} = 0.$$

One explicit choice is the construction above with $p = 3$, $e = 4$, $u(1) = -1, u(2) = 1$, and

$$(c_0, c_1, c_2, c_3) = (1, -9, 27, -27).$$

It satisfies

$$L(s, f) = -(1 - 3^{1-s})^3 L(s, \chi_{-3}), \qquad L'''(1, f) = -\frac{2\pi}{\sqrt{3}}(\log 3)^3 \neq 0.$$

More generally, for any prescribed r, a suitable p^{2^k} supports an odd-integer-valued function whose first r logarithm-weighted harmonic sums vanish.

Proof. Here $u = -\chi_{-3}$, and $C(T) = (1 - 3T)^3$. Use $L(1, \chi_{-3}) = \pi/(3\sqrt{3})$ and differentiate the factor of exact order three. For a zero-mean periodic function, the Dirichlet series and its derivatives converge in the half-plane $\Re s > 0$ after the usual partial summation; in particular the displayed logarithm-weighted series converge and equal the indicated derivatives up to the sign $(-1)^j$. Choose $2^k - 1 \geq r$ for the last assertion. □

Theorem 4.3 (Exact gcd-layer criterion for arbitrary congruence modulus)**.** *Let p be prime, $M \geq 2$ with $\gcd(p, M) = 1$, and let R_M be the rational numbers with denominator coprime to M. Fix $e \geq 1$ and $r \geq 0$. There are coefficients $c_0, \ldots, c_{e-1} \in 1 + MR_M$ for which the function*

$$f(n) = c_{v_p(n)} \quad (v_p(n) < e), \qquad f(n) = 0 \quad (p^e \mid n)$$

has zero mean and a zero of order at least r at $s = 1$ if and only if

$$\sum_{t=j}^{e-1} \binom{t}{j} p^{-(t-j)} \equiv 0 \pmod{MR_M} \quad (0 \leq j \leq r). \tag{4.4}$$

An empty sum is zero, but if $r \geq e - 1$ the test with $j = e - 1$ is one and hence prevents existence. The coefficients can be chosen to be integers with the same congruences.

Proof. For $\Re s > 1$,

$$L(s, f) = (1 - p^{-s})\zeta(s)C(p^{-s}), \qquad C(T) = \sum_{t=0}^{e-1} c_t T^t.$$

The pole of ζ has order one and the other Euler factor is nonzero at one. Thus zero mean and zero order at least r are equivalent to divisibility of C by $(T - 1/p)^{r+1}$. Because $C \equiv S_e \pmod{M}$, this implies the vanishing of the first $r + 1$ Hasse derivatives of S_e at $1/p$, which is precisely (4.4).

Conversely subtract from $S_e(T)$ its Taylor polynomial of degree r at $1/p$. Under (4.4), every coefficient of the Taylor polynomial being subtracted lies in MR_M, so the resulting polynomial C is congruent to S_e and is divisible by $(T - 1/p)^{r+1}$. Its coefficients have denominators prime to M. Multiply by a common denominator chosen to be one modulo M to obtain integer coefficients, preserving the congruences and the zero order. □

Corollary 4.4 (Closed form for prime congruence moduli)**.** *Take $M = \ell \neq p$ prime and put $a = \mathrm{ord}_\ell(p)$ and $D = \ell^{v_\ell(e)}$. The largest root multiplicity at $T = 1/p$ attainable by a polynomial $C \equiv S_e \pmod{\ell}$ is*

$$\mu = \begin{cases} 0, & a \nmid e, \\ D, & a \mid e,\ a > 1, \\ D - 1, & a = 1. \end{cases}$$

Accordingly, a convergent gcd-layer series exists exactly when $\mu \geq 1$, and its maximal zero order at one is $\mu - 1$. In particular, for odd p and odd coefficients, the maximal order in this gcd-layer class is $2^{v_2(e)} - 2$, when a convergent series exists.

Proof. In characteristic ℓ, write $e = Dm$ with $\ell \nmid m$, and factor $T^e - 1 = (T^m - 1)^D$. The point p^{-1} is a root precisely when $a \mid e$. Its multiplicity in $T^e - 1$ is then D, with one removed by division by $T - 1$ exactly when $a = 1$. For attainability, divide S_e in $\mathbb{F}_\ell[T]$ by the maximal power of $1 - \overline{p}T$, lift the quotient to an integer polynomial, and multiply by $(1 - pT)^\mu$. The lifted quotient is nonzero at $1/p$, as its reduction is nonzero there. Hence the multiplicity is exactly μ. □

Corollary 4.5 (A height-optimal double zero at period 64)**.** *Define $f(n) = c_{v_2(n)}$ when $v_2(n) < 6$, and zero when $64 \mid n$, using*

$$(c_0, c_1, c_2, c_3, c_4, c_5) = (1, -5, 7, -2, 4, -8).$$

All its off-zero values are one modulo three, its fundamental period is 64, and

$$L(s, f) = (1 - 2^{-3s})\zeta(s)(1 - 2^{1-s})^3.$$

Consequently

$$L(1, f) = L'(1, f) = 0, \qquad L''(1, f) = \frac{7}{4}(\log 2)^3 \neq 0.$$

Its height $\max_n |f(n)| = 8$ is the least possible among integer gcd-layer functions of period 64, congruent to one modulo three off zero, whose Dirichlet series vanishes to order at least two at one. The displayed coefficient vector is the unique minimizer in that class.

Proof. The coefficient polynomial factors as

$$1 - 5T + 7T^2 - 2T^3 + 4T^4 - 8T^5 = (1 - 2T)^3(1 + T + T^2).$$

Use the identity in Theorem 4.3 and $(1 - T)(1 + T + T^2) = 1 - T^3$. The factor $1 - 2^{1-s}$ has a simple zero with leading coefficient $\log 2$, while $\zeta(s)$ has residue one. The coefficient of $(s - 1)^2$ is therefore $(7/8)(\log 2)^3$, giving the stated second derivative.

For optimality, any competing integer polynomial of degree five must have the factor $(1 - 2T)^3$. That factor is primitive in $\mathbb{Z}[T]$, so Gauss's lemma makes its quotient an integer polynomial. Its leading coefficient c_5 is a nonzero multiple of eight; it cannot be zero since $c_5 \equiv 1 \pmod 3$. The height is thus at least eight, and the displayed example attains it.

For uniqueness at height eight, write $C(T) = (1 - 2T)^3(A + BT + ET^2)$, where $A, B, E \in \mathbb{Z}$. The conditions on $c_5 = -8E$ force $E = 1$. Then $c_4 = 12 - 8B$, its height bound and its congruence force $B = 1$. Finally $c_0 = A \equiv 1 \pmod 3$ and $|c_3| = |6 - 8A| \leq 8$ force $A = 1$. This gives exactly the stated vector. □

These two prime-power theorems expose a genuine distinction. The nonconstant zero-mean layer seed in Theorem 4.1 removes the zeta pole before the coefficient polynomial is used. Constant gcd layers must spend one additional root on that pole. The factor-of-two discrepancy between an arbitrary odd Okada vector and a gcd-invariant one at small prime-power exponents is therefore an exact algebraic effect. The constructions have odd or prescribed-congruence coefficients, not coefficients restricted to $\{\pm 1\}$; they do not solve the remaining square-modulus sign problem.

5 Quadratic weighted binary expansions

The valuation argument of corpus [112] concerns the weight n in a binary expansion. The following extension concerns the weight n^2. All valuation and representation arguments needed here are included. The analytic consequence uses the classical Pólya–Carlson theorem, also used in corpus [088].

For binary digits $\varepsilon_n \in \{0,1\}$, put

$$X_2(\varepsilon) = \sum_{n\ge 1} \frac{n^2 \varepsilon_n}{2^n}.$$

For an odd prime p, write

$$t_p = \operatorname{ord}_p(2), \qquad w_p = v_p(2^{t_p}-1) \ge 1.$$

Theorem 5.1 (Exact quadratic denominator amplification)**.** *Suppose the digits of X_2 have a nonconstant periodic tail, beginning after position N, of least period r. Let $d>1$ be the reduced denominator of the ordinary binary value of this periodic tail, and let Q be the odd part of the reduced denominator of X_2. If $p \mid d$ and*

$$a = v_p(d) > b = v_p(r),$$

then

$$\boxed{v_p(Q) = a + 2w_p.}$$

In particular, Q is divisible by the cube of some odd prime.

Proof. Write the periodic word as $e_1,\dots,e_r$, and set

$$M = 2^r - 1, \quad S = \sum_{i=1}^{r} e_i 2^{r-i}, \quad T = \sum_{i=1}^{r} i e_i 2^{r-i}, \quad U = \sum_{i=1}^{r} i^2 e_i 2^{r-i}.$$

Summing over the positions $i + kr$ gives the three tail values

$$\begin{aligned} y &= \sum_{i\ge 1} \frac{e_i}{2^i} = \frac{S}{M}, \\ z &= \sum_{i\ge 1} \frac{i e_i}{2^i} = \frac{T}{M} + \frac{rS}{M^2}, \\ w &= \sum_{i\ge 1} \frac{i^2 e_i}{2^i} = \frac{U}{M} + \frac{2rT}{M^2} + \frac{r^2 S(M+2)}{M^3}. \end{aligned}$$

Here the digits on the left are extended periodically. The last formula follows from $\sum_{k\ge 0} k^2 q^k = q(1+q)/(1-q)^3$ with $q = 2^{-r}$. The prefix contributes an integer I after multiplication by 2^N, so

$$2^N X_2 = I + N^2 y + 2Nz + w. \tag{5.1}$$

The ordinary radix-expansion theorem gives $r = \operatorname{ord}_d(2)$. Put $k = v_p(M)$. The lifting formula at an odd prime gives $k = w_p + b$, while reduction of S/M gives $v_p(S) = k - a$. Since $M+2$ is a p-adic unit, the last term in w has exact valuation

$$2b + (k-a) - 3k = -a - 2w_p.$$

The middle term has valuation at least $b - 2k = -b - 2w_p$, and the first at least $-k = -b - w_p$. Both bounds are strictly larger because $a > b$. Hence $v_p(w) = -a - 2w_p$. Similarly, the displayed formula for z gives $v_p(z) = -a - w_p$, while $v_p(y) = -a$. Thus no other term in (5.1) can cancel the term of least valuation. This proves the exact formula for Q.

To obtain the final assertion, choose the largest prime $p \mid d$. For an odd prime q and an integer $c \ge 1$, the order formula is

$$\operatorname{ord}_{q^c}(2) = t_q q^{\max(0, c - w_q)}.$$

Taking the least common multiple of these orders over $q \mid d$ shows that $b = \max(0, a - w_p) < a$: a smaller prime $q < p$ contributes no factor p, because $t_q \mid q - 1 < p$. The exact formula therefore gives $v_p(Q) = a + 2w_p \geq 3$. □

Corollary 5.2 (Aperiodicity for cube-free odd denominators)**.** *Every $x \in [0, 6]$ has a representation*

$$x = \sum_{n \geq 1} \frac{n^2 \varepsilon_n}{2^n}, \qquad \varepsilon_n \in \{0, 1\}.$$

If x is rational and the odd part $Q > 1$ of its reduced denominator is cube-free, then every such representation is aperiodic after every finite prefix. This applies, for example, to both $1/3$ and $1/9$. The prime-cube obstruction is sharp: the periodic word 10 gives

$$\sum_{k \geq 0} \frac{(2k+1)^2}{2^{2k+1}} = \frac{82}{27}.$$

Proof. Let $u_n = n^2/2^n$. Its future tail is

$$T_n = \sum_{j > n} u_j = \frac{n^2 + 4n + 6}{2^n},$$

so $u_n \leq T_n$ for every n and the total sum is 6. Given a remainder in $[0, u_n + T_n]$, choose the next digit so that the new remainder lies in $[0, T_n]$. This is possible because $[0, T_n] \cup [u_n, u_n + T_n] = [0, u_n + T_n]$. Since $T_n \to 0$, the construction proves existence for every $x \in [0, 6]$.

A nonconstant periodic tail contradicts Theorem 5.1. An eventually zero tail gives a dyadic rational, and an eventually one tail does too, by the formula for T_n. Neither is possible when $Q > 1$. Finally, the word 10 has $r = 2$, $M = 3$, and $S = T = U = 2$, so its quadratic weighted value is $2/3 + 8/9 + 40/27 = 82/27$. □

Theorem 5.3 (Finite bound on all eventual periods)**.** *For an odd positive integer Q, let $\mathcal{P}_Q^{(2)}$ be the smallest set of odd primes containing every p with $v_p(Q) > 2w_p$, and containing every odd prime divisor of t_q whenever it contains q. This set is finite. Define*

$$e_p = \max \left\{ 0,\ v_p(Q) - 2w_p,\ \max_{\substack{q \in \mathcal{P}_Q^{(2)} \\ q > p}} v_p(t_q) \right\}, \qquad D_Q^{(2)} = \prod_{p \in \mathcal{P}_Q^{(2)}} p^{e_p},$$

with empty maxima and products interpreted as 0 and 1. For every nonconstant eventual tail of a quadratic weighted binary representation with odd denominator Q, its ordinary binary denominator d and least period r satisfy

$$d \mid D_Q^{(2)}, \qquad r \mid \operatorname{ord}_{D_Q^{(2)}}(2).$$

In particular, such a tail requires $D_Q^{(2)} > 1$.

Proof. Every newly added prime is smaller than its parent because $t_q \mid q - 1$, proving finiteness. Write $a_p = v_p(d)$ and $b_p = v_p(r)$. If $a_p > b_p$, Theorem 5.1 gives $a_p = v_p(Q) - 2w_p$. Otherwise $a_p \leq b_p$, and the order formula in the preceding proof gives

$$b_p = \max \left\{ \max(0, a_p - w_p),\ \max_{\substack{q \mid d \\ q > p}} v_p(t_q) \right\}.$$

Its first entry is strictly smaller than a_p. Thus a larger $q \mid d$ satisfies $a_p \leq v_p(t_q)$. Process the prime factors of d in decreasing order. Each is either an initial seed with the stated exponent bound, or is introduced by an already processed larger prime and again has exponent at most e_p. This proves $d \mid D_Q^{(2)}$. Divisibility of multiplicative orders gives the assertion about r. □

This is a necessary period bound, not a sufficient condition for the existence of a periodic representation of a specified numerator. For $Q = 27$, it gives $D_Q^{(2)} = 3$ and $r \mid 2$. For a fully finite example, trial division through 31 proves that 1093 is prime, and exact modular exponentiation gives

m	364	182	52	28
$2^m \bmod 1093$	1	1092	27	121.

Since the prime divisors of 364 are $2, 7, 13$, these congruences prove $\mathrm{ord}_{1093}(2) = 364$. Also

$$2^{364} \equiv 1 \pmod{1093^2}, \qquad 2^{364} \equiv 1064432260 \not\equiv 1 \pmod{1093^3},$$

so $w_{1093} = 2$. These are the finite checks recorded in [112]; the two prime-power congruences were recomputed for this manuscript. They give $D_{1093^4}^{(2)} = 1$. Thus every quadratic weighted binary representation of $1/1093^4$ is aperiodic. This is a consequence for one explicitly checked prime, and makes no assertion about the infinitude of Wieferich primes.

Corollary 5.4 (Rational derivative data and a natural boundary)**.** *For binary digits define*

$$Y = \sum_{n\geq 1} \frac{\varepsilon_n}{2^n}, \qquad F(z) = \sum_{n\geq 1} \varepsilon_n z^n.$$

The following conditions are equivalent: the digits are eventually periodic; Y is rational; F is a rational function. If x satisfies Corollary 5.2, then every quadratic weighted representation of x therefore gives

$$Y \notin \mathbb{Q}, \qquad F \text{ has the unit circle as a natural boundary}, \qquad \left(z\frac{d}{dz}\right)^2 F(z)\Big|_{z=1/2} = x \in \mathbb{Q}.$$

In particular, F is transcendental over $\mathbb{C}(z)$.

Proof. The equivalence between eventual periodicity and rationality of Y is the ordinary radix-expansion theorem. Periodicity makes F rational by the geometric-series formula. Conversely, a rational F gives an eventual linear recurrence with constant rational coefficients. A state consisting of sufficiently many consecutive digits determines the next digit. There are only finitely many such binary states, so a repeated state forces eventual periodicity.

For a nonperiodic sequence, infinitely many digits are one, and the integer-coefficient series F has radius of convergence exactly one. The Pólya–Carlson theorem [5] says that such a series is either rational or has the unit circle as a natural boundary. The rational case has been excluded. An algebraic function has only finitely many possible singularities and branch points, so cannot have that natural boundary. The derivative identity follows by termwise differentiation strictly inside the circle of convergence. □

Corollary 5.5 (An exact recurrence criterion for binary sequences)**.** *For a binary sequence ε_n, the following six conditions are equivalent:*

(i) *the sequence is eventually periodic;*
(ii) *$\sum_{n\geq 1} \varepsilon_n 2^{-n}$ is rational;*
(iii) *$F(z) = \sum_{n\geq 1} \varepsilon_n z^n$ is rational;*
(iv) *$F(z)$ is algebraic over $\mathbb{C}(z)$;*
(v) *$F(z)$ is D-finite over $\mathbb{C}(z)$;*
(vi) *the sequence is P-recursive: for some polynomials $P_0, \ldots, P_m \in \mathbb{C}[n]$, not all zero,*

$$\sum_{j=0}^{m} P_j(n)\varepsilon_{n+j} = 0$$

for all sufficiently large n.

Consequently every quadratic weighted representation in Corollary 5.2 fails all six conditions, including every nonzero polynomial-coefficient linear recurrence.

Proof. The first three equivalences were proved in Corollary 5.4. A rational function is algebraic and D-finite. An algebraic function cannot have a full circle as a natural boundary, since its possible branch points and singularities belong to a finite set. A D-finite analytic germ satisfies a nonzero linear differential equation with polynomial coefficients, and can be continued along paths avoiding the finite zero set of its leading coefficient. It therefore cannot have the unit circle as a natural boundary either. The Pólya–Carlson dichotomy in the preceding proof forces rationality in both cases.

Finally, coefficient extraction converts a polynomial-coefficient differential equation into an eventual polynomial-coefficient recurrence, and summing such a recurrence converts it into a polynomial-coefficient differential equation with at most a polynomial inhomogeneous term. Applying sufficiently many derivatives removes that term. This proves the equivalence of D-finiteness and P-recursiveness used here. □

A further standard consequence links this criterion to corpus [111]: no such digit generating function can be a diagonal of a rational function over $\mathbb{Q}$ in finitely many variables, since these diagonals are D-finite in characteristic zero. This is a consequence of the classical diagonal theorem, not a separate conjecture resolution.

6 The complete binomial gcd distribution in a prime family

Corpus [086] proves a minimum-value criterion by examining the base-prime digits of three nearby primes. Its increasing-gap half also yields the full distribution of the binomial gcd values.

Theorem 6.1 (Exact four-value histogram)**.** *Let $p < q < r$ be primes and put*

$$a = q - p, \quad c = r - q, \quad b = a + c = r - p, \quad n = pqr.$$

Suppose $2b^2 < p$ and $c > a$. Set

$$\begin{aligned} A_p &= 2(a+b+1)(ab+1) - 2, \\ A_q &= 2(c-a)(q-ac+1) - 2, \\ A_r &= (r-b-c+1)(bc+1) - 2. \end{aligned}$$

For $1 \le k \le n-1$, the distribution is exactly

$\gcd(n, \binom{n}{k})$	*number of indices*
qr	A_p
pr	A_q
pq	A_r
n	$n - 1 - A_p - A_q - A_r$.

For $1 < k \le n/2$, the first three counts are halved, and the last count is $(n - 3 - A_p - A_q - A_r)/2$. Furthermore, uniformly in this family, all but $O(pb^2)$ of the interior coefficients are divisible by the entire integer n. The exceptional proportion is $O(b^2/p^2) = O(1/p)$.

Proof. We include the pair-exclusion argument from [086] so the arithmetic input is explicit. By symmetry it is enough to consider $1 \le k \le n/2$. If two distinct primes $s, t \mid n$ fail to divide $\binom{n}{k}$, Lucas' lowest digit forces $k = stj$, where $1 \le j \le (u-1)/2$ and u is the remaining prime. The necessary conditions are that tj is digitwise at most tu in base s, and sj is digitwise at most su in base t. The product expansions give

$$qr = (1, a+b, ab)_p, \qquad pr = (1, c-a-1, q-ac)_q, \qquad pq = (r-b-c, bc)_r.$$

All displayed digits are in range under $2b^2 < p$. We will use $a(a+b), b(a+b), c(b+c) < 2b^2 < p$.

For the pair $\{p, q\}$, write $qj = (j + \lfloor aj/p \rfloor)p + (aj \bmod p)$. Since $j < (p+b)/2$ and $b + 2a < p$, its middle digit is less than p. The middle-digit bound gives $j \le a + b$, hence $aj < p$. The last-digit bound now gives $aj \le ab$, so $j \le b$. In base q we have $pj = (j-1)q + (q - aj)$. Its two digit bounds force respectively $j \le c - a$ and $j \ge c$, a contradiction.

For the pair $\{p, r\}$, similarly $rj = (j + \lfloor bj/p \rfloor)p + (bj \bmod p)$ has two digits, since $j < (p+a)/2$ and $a + 2b < p$. The middle-digit bound gives $j \le a + b$, so $bj < p$; the last-digit bound then yields $bj \le ab$ and $j \le a$. In base r, $pj = (j-1)r + (r - bj)$, and the last-digit bound requires $r \le b(j+c) \le b^2 < p$, again impossible.

For $\{q, r\}$, write $rj = (j + \lfloor cj/q \rfloor)q + (cj \bmod q)$. Here $j < p/2$ makes the middle digit less than q, so its bound gives $j \le c - a - 1 < c$. Thus $cj < c^2 < r$, and in base r, $qj = (j-1)r + (r - cj)$. Its last-digit bound requires $r \le c(j+b) < c(b+c) < p$, the final contradiction. Therefore at least two primes divide every interior coefficient.

For a prime s and base-s digits n_j of n, Lucas' theorem gives

$$\#\{0 \le k \le n : s \nmid \tbinom{n}{k}\} = \prod_j (n_j + 1).$$

Indeed each digit of k can independently be chosen from 0 to n_j. Excluding $k = 0, n$ subtracts two. The product expansions give the base-prime digits

$$qr = (1, a+b, ab)_p, \qquad pr = (1, c-a-1, q-ac)_q, \qquad pq = (r-b-c, bc)_r.$$

Every displayed digit is in range because $2b^2 < p$. Multiplication by the corresponding missing prime appends a zero digit to obtain n. The Lucas counts are therefore A_p, A_q, A_r. The three nondivisibility sets are disjoint by the first paragraph, so these are exactly the counts for gcd $n/p, n/q, n/r$; the remaining indices have gcd n.

Since n is odd, the symmetry $k \leftrightarrow n - k$ halves each interior count. Excluding $k = 1$ removes one only from the gcd-n count. Finally the three explicit expressions are respectively $O(b^3)$, $O(pb)$, and $O(pb^2)$. As $b < p$ and $n \ge p^3$, the claimed concentration estimates follow. □

For $p = 101$, $q = 103$, and $r = 107$, the hypotheses hold because $2 \cdot 6^2 = 72 < 101$ and $4 > 2$. Here $n = 1,113,121$, and for $2 \le k \le 556,560$ the four counts are

$\gcd(n, \binom{n}{k})$	count
10,403	1,224
10,807	191
11,021	116
1,113,121	555,028.

They sum to 556,559. An independent computation using Legendre's factorial valuation formula checked every index in this interval. This histogram extends the conclusion of [086]; it does not classify arbitrary squarefree integers or settle the remaining asymptotic upper-bound question in Erdős Problem 700.

7 Translated reciprocal sums in one number field

The rational multilinear-tail argument of corpus [080] admits a number field extension. The new input is the norm of a nonzero algebraic integer; no sign assumption is imposed on its conjugates.

Theorem 7.1 (Growth versus common field degree)**.** *Let $a_1 < a_2 < \cdots$ be positive integers satisfying*

$$\sup_n a_n \sum_{k \ge n} \frac{1}{a_k} < \infty. \tag{7.1}$$

For rational h avoiding all poles $-a_n$, write

$$S(h) = \sum_{n\geq 1} \frac{1}{a_n + h}.$$

Suppose that, for d distinct rational numbers $h_1, \ldots, h_d$, all $S(h_i)$ belong to a number field K of degree $D = [K : \mathbb{Q}]$. Then

$$\boxed{\log a_n = O\big((dD+1)^{n/d}\big).}$$

For integer shifts, after a finite truncation, there is a constant C such that

$$\sum_{j=0}^{d-1} \log a_{n+j} \leq dD \sum_{k<n} \log a_k + C.$$

Proof. First take integer shifts $h_1 < \cdots < h_d$, and discard finitely many terms so all denominators are positive. The resulting tail values are denoted by $S(h_i)$ again; each differs from its original value by a rational number and hence still belongs to K. Write

$$\alpha_k^{(i)} = a_k + h_i, \quad \Pi_n^{(i)} = \prod_{k<n} \alpha_k^{(i)}, \quad R_n^{(i)} = S(h_i) - \sum_{k<n} \frac{1}{\alpha_k^{(i)}}.$$

Choose a positive integer Q for which every $QS(h_i)$ is an algebraic integer. Then $Q\Pi_n^{(i)} R_n^{(i)}$ is an algebraic integer. Define the strictly positive sum

$$O_n = \sum_{\substack{k_1,\ldots,k_d \geq n \\ k_1,\ldots,k_d \text{ distinct}}} \prod_{i=1}^{d} \frac{1}{\alpha_{k_i}^{(i)}}.$$

For a block $B \subseteq \{1, \ldots, d\}$ put $c_{B,i} = \prod_{j\in B\setminus\{i\}} (h_j - h_i)^{-1}$. Partial fractions give

$$\prod_{i\in B} \frac{1}{x + h_i} = \sum_{i\in B} \frac{c_{B,i}}{x + h_i}.$$

For a set partition π, let $\mu(\pi) = \prod_{B\in\pi} (-1)^{|B|-1}(|B|-1)!$. Möbius inversion on set partitions, followed by the displayed partial fractions, gives the exact identity

$$O_n = \sum_{\pi} \mu(\pi) \prod_{B\in\pi} \left(\sum_{i\in B} c_{B,i} R_n^{(i)} \right). \tag{7.2}$$

The inversion removes precisely the tuples having repeated indices; all sums converge absolutely.

Let $\Delta = \prod_{i<j} (h_j - h_i) > 0$. In an expanded term of (7.2), the representatives chosen from different blocks are distinct. The denominator of the product of their $c_{B,i}$ coefficients is a product over distinct unordered pairs and divides Δ. Consequently

$$Z_n = M_n O_n, \qquad M_n = \Delta Q^d \prod_{i=1}^{d} \Pi_n^{(i)},$$

is a nonzero algebraic integer.

For every embedding $\sigma : K \longrightarrow \mathbb{C}$,

$$\sigma(R_n^{(i)}) = \sigma(S(h_i)) - \sum_{k<n} \frac{1}{\alpha_k^{(i)}}$$

is uniformly bounded in n. Equation (7.2) therefore gives $|\sigma(O_n)| \le B$ for a fixed $B \ge 1$, simultaneously for all embeddings and all n. The norm of Z_n is a nonzero rational integer. At the distinguished real embedding $O_n > 0$, and thus

$$1 \le \left|\mathrm{Norm}_{K/\mathbb{Q}}(Z_n)\right| = M_n^D O_n \prod_{\sigma \ne \mathrm{id}} |\sigma(O_n)| \le M_n^D B^{D-1} O_n.$$

Put $P_n = \sum_{k<n} \log a_k$. The convergence of $\sum 1/a_k$ implies that the partial products for every fixed shift differ from $\prod_{k<n} a_k$ by uniformly bounded positive factors. Hence $M_n \le C_1 e^{dP_n}$. Symmetrization and then removal of the ordering restrictions on distinct indices give

$$O_n \le d! \prod_{j=0}^{d-1} \left(\sum_{k \ge n+j} \frac{1}{a_k + h_1} \right) \le C_2 \prod_{j=0}^{d-1} a_{n+j}^{-1},$$

where the last inequality is (7.1). Combining the estimates and taking logarithms proves

$$P_{n+d} \le (dD+1)P_n + C.$$

Iteration on each residue class modulo d gives the required bound.

For rational shifts take a common denominator w, write $h_i = u_i/w$, and replace a_n by $b_n = w a_n$. The translated sums become $S(h_i)/w \in K$, condition (7.1) is unchanged, and $\log b_n = \log a_n + O(1)$. The integer-shift result applies. □

Corollary 7.2 (Finitely many translated values in each fixed field)**.** *Under* (7.1)*, suppose*

$$\lambda = \limsup_{n \to \infty} (\log a_n)^{1/n} > 1.$$

If d distinct translated values $S(h_i)$ lie in a number field of degree D, then

$$\lambda^d \le dD + 1.$$

Thus each fixed number field contains only finitely many values $S(h)$ at rational shifts avoiding the poles. If $1 < \lambda < \infty$ and d selected values are all algebraic, then

$$\boxed{[\mathbb{Q}(S(h_1), \dots, S(h_d)) : \mathbb{Q}] \ge \left\lceil \frac{\lambda^d - 1}{d} \right\rceil.}$$

If $\lambda = \infty$, every allowed rational-shift value $S(h)$ is transcendental.

Proof. The growth bound of Theorem 7.1 gives $\lambda \le (dD+1)^{1/d}$. For fixed D, this fails for all sufficiently large d, proving finiteness. Apply the same inequality to the field generated by the selected algebraic values. When $\lambda = \infty$, apply the theorem with $d = 1$ to the number field generated by any putatively algebraic value; the resulting finite bound $\lambda \le D + 1$ is impossible. □

The conclusion applies uniformly to every positive integer sequence with $a_n/2^{2^n} \to 1$, the perturbation class in Erdős Problem 263 [7]. Indeed $a_{n+1}/a_n \to \infty$, so eventual monotonicity and the tail hypothesis follow after discarding a finite prefix, while $\log a_n = 2^n \log 2 + o(1)$ gives $\lambda = 2$. If all values at d distinct rational shifts are algebraic, their compositum has degree at least $\lceil (2^d - 1)/d \rceil$. For $d = 5, 10, 16$ these lower bounds are $7, 103, 4096$, respectively. At most one allowed shift value belongs to $\mathbb{Q}$, and at most two belong to any specified quadratic field. In particular no three distinct rational shifts can all give values in the same quadratic field, for any sequence in this entire perturbation class.

An explicit perturbed example is $a_n = 2^{2^n} + n$. It has $a_{n+1} \ge 3a_n$ and hence tail constant at most $3/2$. These are common-field restrictions; they do not determine the arithmetic nature of a specified $S(0)$ throughout this class or settle Erdős Problems 263 or 265. We deliberately do not label the

unperturbed Mahler/Fermat special values as newly unresolved: classical transcendence results already apply to important instances of those values [8].

For comparison, $a_n = 2^{n!}$ satisfies the tail hypothesis and has $\lambda = \infty$. Thus every number $\sum_{n\geq 1}(2^{n!} + h)^{-1}$ at an allowed rational shift h is transcendental. This is the classical rapid-growth mechanism; the fixed-field degree restriction above reaches the slower sequence 2^{2^n} where that one-variable conclusion does not follow.

Checks. Exact rational arithmetic verified the quadratic valuation identity in $4,330$ prime-valuation cases coming from primitive binary periods at most eight and prefix lengths from zero through four. The binomial histogram example was independently verified at every relevant index using Legendre's formula, rather than the Lucas count used in the proof. These checks supplement the all-index arguments above.

8 A geometric converse to polynomial primitive towers

Source and additional step. Corpus [113] proves that polynomial covers possess algebraic primitive towers, and [114] organizes stability by homology and monodromy. We extract a converse that does not need the representation-theoretic classification: universal integrability detects the genus and the number of points above infinity. Its proof uses trace descent, approximation of finite jets, and the classical de Rham theorem for affine curves. These are structural consequences of classical tools; no assertion of historical priority is made.

Let X be a smooth connected projective complex curve, let $x \in K = \mathbb{C}(X)$ be nonconstant, and let S be the nonempty set of poles of x. Put $U = X \setminus S$, $A = \mathcal{O}(U)$, and $D = d/dx$. Write g for the genus of X and $s = |S|$. All assertions concern characteristic zero. An algebraic primitive is allowed initially in a finite algebraic extension of K; elementary primitives are allowed in elementary extensions with the same constants.

Lemma 8.1 (Trace descent and absence of new finite poles)**.** *If $f \in K$ has an algebraic nth primitive, it has one in K. If $f \in A$ and $Dh = f$ for $h \in K$, then $h \in A$.*

Proof. For a finite extension L/K, the derivation extends uniquely and commutes with all K-embeddings. Consequently it commutes with trace. If $D^n h = f$, then $D^n(\mathrm{Tr}_{L/K}(h)/[L : K]) = f$. For the second assertion, $dh = f\,dx$ has no pole on U. A pole of order $m > 0$ of h would give a pole of order $m + 1$ of dh, which is impossible. □

Theorem 8.2 (Exact dimension of the algebraic-integration obstruction)**.** *Let*

$$I_x = \{f \in A : f \text{ has an algebraic primitive with respect to } x\}.$$

Then

$$\boxed{\dim_{\mathbb{C}}(A/I_x) = 2g + s - 1.}$$

More precisely, $f \mapsto [f\,dx]$ induces an isomorphism $A/I_x \simeq H^1_{\mathrm{dR}}(U)$.

Proof. Because U is a smooth affine curve, $H^1_{\mathrm{dR}}(U) = \Omega^1_A/dA$. The kernel of the indicated map is I_x, by Lemma 8.1. To prove surjectivity, take $\omega \in \Omega^1_A$. The zeros of dx on U form a finite set. At such a point P, choose a local parameter z and put $m_P = \mathrm{ord}_P(dx) \geq 1$. Choose a truncated Taylor primitive h_P of ω through degree m_P, so that $\omega - dh_P$ vanishes to order at least m_P. The Chinese remainder theorem in A supplies an $h \in A$ with $h - h_P \in \mathfrak{m}_P^{m_P+1}$ at every chosen point. Therefore $(\omega - dh)/dx$ has no pole anywhere on U, and equals some $f \in A$. Thus $[\omega] = [f\,dx]$. The classical algebraic de Rham comparison and the topology of a genus-g surface with s punctures give $\dim H^1_{\mathrm{dR}}(U) = 2g + s - 1$. □

Theorem 8.3 (Universal algebraic stability characterizes polynomial covers)**.** *The following statements are equivalent:*

1. *Every $f \in A$ has an algebraic primitive.*
2. *Every $f \in A$ has a coherent infinite tower of algebraic primitives.*
3. *$g = 0$ and $s = 1$.*
4. *There is a coordinate t with $K = \mathbb{C}(t)$ and $x = P(t)$ for a nonconstant polynomial $P \in \mathbb{C}[t]$.*

In the affirmative case every tower can be chosen in K.

Proof. The second assertion implies the first. Theorem 8.2 makes the first equivalent to $2g+s-1 = 0$, hence to $g = 0, s = 1$. A genus-zero smooth projective curve over $\mathbb{C}$ is the projective line. Choosing its unique point in S as infinity gives $A = \mathbb{C}[t]$ and makes x a polynomial in t; the converse is immediate. Finally, on a polynomial cover define

$$J_P Q(t) = \int_0^t P'(u)Q(u)\,du, \qquad Q \in \mathbb{C}[t].$$

This is polynomial and $D(J_P Q) = Q$. Hence $J_P^n Q$ gives a coherent infinite tower for every $Q \in A$. □

Theorem 8.4 (Universal elementary stability characterizes genus zero)**.** *The following are equivalent:*

1. *Every $f \in K$ has an elementary primitive with respect to x.*
2. *Every $f \in K$ has elementary primitives of every order.*
3. *$g = 0$.*

When $g > 0$, the g-dimensional space $\{\omega/dx : \omega \in H^0(X, \Omega^1_X)\}$ consists, except for zero, entirely of obstructions already at the first integration.

Proof. If $g = 0$, write $K = \mathbb{C}(t)$ and $x = R(t)$ with R rational. For every $j \geq 0$, the differential $x^j f\,dx$ is rational in t, and hence has an elementary primitive H_j by partial fractions. For $n \geq 1$, repeated integration by parts gives

$$F_n = \frac{1}{(n-1)!} \sum_{j=0}^{n-1} (-1)^j \binom{n-1}{j} x^{n-1-j} H_j, \qquad D^n F_n = f.$$

Using the same chosen primitives H_j for all n gives $DF_n = F_{n-1}$ for $n \geq 2$ directly, so these form a coherent tower.

Conversely, suppose $g > 0$ and choose a nonzero holomorphic differential ω. Put $f = \omega/dx \in K$. Liouville's theorem for integration of algebraic functions, followed by trace and norm from its finite algebraic extension, would express an elementary-integrable ω as

$$\omega = dh + \sum_{i=1}^{r} c_i\,du_i/u_i, \qquad h \in K, \quad u_i \in K^*, \quad c_i \in \mathbb{C}.$$

A pole of h would produce a pole of order at least two in dh, which the logarithmic differentials cannot cancel. Thus h is constant. The vanishing residues of ω then give $\sum_i c_i \operatorname{ord}_P(u_i) = 0$ for all P. The finite matrix of these valuations has integer entries, so its complex kernel is generated by rational vectors. Multiplying a rational kernel vector by a common denominator yields integers a_i with $\operatorname{div}(\prod_i u_i^{a_i}) = 0$. The product is constant on X, and hence $\sum_i a_i\,du_i/u_i = 0$. Every residue-free linear combination in the display is therefore zero, contrary to $\omega \neq 0$. This proves both necessity and the last statement. □

An explicit elliptic obstruction. For $y^2 = x^3 - x$, the smooth projective curve has genus one and one point above infinity. Therefore $A = \mathbb{C}[x, y]/(y^2 - x^3 + x)$ has exactly two independent one-step algebraic-integration obstructions. The identities

$$d((3x^2 - 2)y) = dx/y + (21/2)xy\,dx,$$
$$d(xy) = x\,dx/y + (5/2)y\,dx$$

give concrete representatives. Since dx/y is a nonzero holomorphic differential, xy has no elementary primitive. The classes of dx/y and $x\,dx/y$ are independent: if the second were a multiple of the first plus an exact differential, its primitive would have a unique simple pole at infinity, which is impossible on a genus-one curve by Riemann–Roch. Thus xy and y represent the two independent algebraic obstructions. For $y^2 = P(x)$ with P squarefree of degree $d \geq 2$, the dimension in Theorem 8.2 is $d - 1$. This count is about first primitives; it must not be confused with the full repeated-integration classification of [114].

What this connection establishes. Universal integrability on an entire algebraic function ring or field is equivalent to a precise geometric condition. It does not prove elementary stability of arbitrary transcendental elementary inputs, and it does not certify the broader claims of corpus [119]–[121].

9 A positivity threshold, finite moment tests, and a martingale comparison

This section develops the inverse-Mellin argument of corpus manuscript [116]. The link between complete monotonicity and a Stieltjes moment sequence is already explicit in Simon's paper [3, Section 3]. The additional conclusions here concern the structure of the actual positivity boundary, its finite moment approximations, a parameter evolution equation, and a probabilistic comparison. They do not provide a closed formula for that boundary. The strict separation from the logarithmic infinite-divisibility boundary is the starting result of [116]; the analytic ingredients needed for it are included below.

Fix throughout

$$\gamma > 1, \qquad 0 < \alpha < 1/\gamma, \qquad \nu = 1 - \alpha\gamma \in (0, 1).$$

For $\beta > 0$, write

$$F_\beta(z) = \sum_{n=0}^{\infty} \frac{z^n}{\Gamma(\beta + \alpha n)^\gamma}, \qquad m_n(\beta) = \frac{n!\,\Gamma(\beta)^\gamma}{\Gamma(\beta + \alpha n)^\gamma}.$$

The power series for F_β is entire. Set

$$\beta_{\mathrm{ID}} = \max_{t>0} \left[\alpha + \frac{\alpha}{t} \log\left(\frac{\gamma(1 - e^{-t})}{1 - e^{-t/\alpha}} \right) \right]. \tag{9.1}$$

This is the threshold for the scalar inequality in Simon's conjecture, and is not the threshold for complete monotonicity.

9.1 Analytic preparation

For $\beta \geq \alpha$ and $\Re s > -1$, use the holomorphic logarithm of Gamma in the right half-plane to define

$$M_\beta(s) = \Gamma(1 + s)\Gamma(\beta)^\gamma \exp\{-\gamma \operatorname{Log}\Gamma(\beta + \alpha s)\}.$$

The following real, initially signed, density is well defined:

$$q_\beta(x) = \frac{1}{2\pi i x} \int_{-i\infty}^{i\infty} M_\beta(s) x^{-s}\,ds, \qquad x > 0. \tag{9.2}$$

Lemma 9.1 (Uniform endpoint estimates)**.** *The function $q_\beta(x)$ is jointly real analytic in $\beta > \alpha$ and $x > 0$. It has total signed mass one and*

$$\int_0^\infty x^s q_\beta(x)\,dx = M_\beta(s), \qquad \Re s > -1. \tag{9.3}$$

For every compact $K \subset (\alpha, \infty)$, uniformly for $\beta \in K$,

$$q_\beta(x) = \frac{\Gamma(\beta)^\gamma}{\Gamma(\beta-\alpha)^\gamma} + O_K(x^\eta), \qquad x \downarrow 0, \tag{9.4}$$

$$q_\beta(x) = \frac{C_\beta}{x\sqrt{2\pi\nu}} r^{\kappa_\beta + 1/2} e^{-\nu r}(1 + o_K(1)), \qquad x \to \infty, \tag{9.5}$$

where

$$\kappa_\beta = \frac{1+\gamma}{2} - \gamma\beta, \qquad r = (\alpha^{\alpha\gamma} x)^{1/\nu},$$
$$C_\beta = (2\pi)^{(1-\gamma)/2}\Gamma(\beta)^\gamma \alpha^{\gamma(1/2-\beta)},$$

where any $0 < \eta < \min(1, \min K/\alpha - 1)$ is allowed. In particular, the densities are positive near both endpoints uniformly on K, and their absolute values have every positive exponential moment. At $\beta = \alpha$, the same Mellin identity holds, the upper-end estimate continues to hold, and $q_\alpha(x) = O_\varepsilon(x^{-\varepsilon})$ as $x \downarrow 0$, for every $\varepsilon > 0$.

Proof. Vertical Stirling estimates give a polynomial in $|\Im s|$ times $e^{-\pi\nu|\Im s|/2}$ on each permissible fixed vertical line. This proves absolute convergence, differentiation under the integral, and analyticity in the logarithmic variable $\log x$; small complex neighborhoods of β have the same estimates. Moving the contour to $\Re s = -1 - \eta$ crosses only the numerator pole at $s = -1$ and gives (9.4). When $\beta = \alpha$, moving instead to $\Re s = -1 + \varepsilon$ gives the stated bound without crossing a pole.

For completeness the uniform large-x step is as follows. Stirling's formula on the right half-plane gives

$$M_\beta(s) = C_\beta s^{\kappa_\beta} \exp\{\nu s(\log s - 1) - \alpha\gamma s \log\alpha\}(1 + O_K(|s|^{-1})).$$

Shift to $\Re s = r$, where $\log x + \alpha\gamma\log\alpha = \nu\log r$, and set $s = r + i\sqrt{r}\,u$. The normalized integrand tends to $e^{-\nu u^2/2}$. If $v = u/\sqrt{r}$, its modulus is bounded by

$$D_K(1+v^2)^{\kappa_\beta/2} e^{\nu r H(v)}, \qquad H(v) = \tfrac{1}{2}\log(1+v^2) - v\arctan v.$$

The bounds $H(v) \le -v^2/4$ for $|v| \le 1$ and $H(v) \le H(1) - \pi(|v|-1)/4$ for $|v| \ge 1$ prove uniform dominated convergence on the central part and exponential negligibility of the outer part. This proves (9.5). These are the uniform saddle estimates used in [116], based on [4, Section 5.11].

The endpoint bounds imply integrability of the signed density and of $x^s q_\beta(x)$ for $\Re s > -1$. In the coordinate $y = \log x$, Fourier inversion identifies the transform of $e^y q_\beta(e^y)$ with $M_\beta(iu)$. Analytic continuation in s proves (9.3), including mass one at $s = 0$. Since $1/\nu > 1$, the upper tail also gives all positive exponential moments of $|q_\beta|$. □

For $\beta \ge \alpha$, absolute integration of the exponential series therefore gives

$$\int_0^\infty e^{zx} q_\beta(x)\,dx = \Gamma(\beta)^\gamma F_\beta(z), \qquad z \in \mathbb{C}. \tag{9.6}$$

Bernstein's theorem and uniqueness of Laplace transforms of finite signed measures imply the exact equivalence

$$F_\beta(-x) \text{ is completely monotone} \iff q_\beta(x) \ge 0 \quad (x > 0). \tag{9.7}$$

9.2 A parameter coupling and a single positivity boundary

Lemma 9.2 (Beta factors)**.** *For every $\beta, \delta > 0$, there is a random variable $Z_{\beta,\delta}$ with $0 < Z_{\beta,\delta} < 1$ such that*

$$\mathbb{E}Z_{\beta,\delta}^{s} = \left[\frac{\Gamma(\beta+\delta)\Gamma(\beta+\alpha s)}{\Gamma(\beta)\Gamma(\beta+\delta+\alpha s)}\right]^{\gamma}, \qquad \Re s \geq 0. \tag{9.8}$$

Consequently the complete-monotonicity property, once true at β, is true at every larger parameter. When both parameters belong to the positive-density region, they can be coupled by

$$X_{\beta+\delta} \overset{d}{=} X_\beta Z_{\beta,\delta}, \qquad X_\beta \text{ independent of } Z_{\beta,\delta}. \tag{9.9}$$

Proof. The logarithm of the right side of (9.8) is

$$\gamma \int_0^\infty (e^{-\alpha s t} - 1)\frac{e^{-\beta t}(1-e^{-\delta t})}{t(1-e^{-t})}\, dt.$$

Its positive kernel is a subordinator Levy measure: near zero it is $\gamma\delta/t + O(1)$, and it decays exponentially at infinity. Exponentiating the negative of that finite-valued subordinator defines Z. Equivalently, it is the multiplicative convolution power γ of B^α, where B has the beta law $\mathrm{Beta}(\beta, \delta)$. Its law has no atom at one. The gamma-ratio identity gives $m_n(\beta+\delta) = m_n(\beta)\mathbb{E}Z^n$.

If complete monotonicity holds at β, Bernstein's law has moments $m_n(\beta)$. The product with an independent Z has moments $m_n(\beta+\delta)$. The entire series for $F_{\beta+\delta}$ shows by monotone convergence that this product has all positive exponential moments and moment generating function $\Gamma(\beta+\delta)^\gamma F_{\beta+\delta}$. Thus its Laplace transform proves complete monotonicity at $\beta+\delta$. Moment or Laplace-transform uniqueness also proves (9.9). □

Theorem 9.3 (One true threshold, with strict interior positivity)**.** *There is a unique number $\beta_c = \beta_c(\alpha, \gamma)$ such that*

$$\alpha < \beta_c < \beta_{\mathrm{ID}} < \frac{\alpha(1+\gamma)}{2}, \tag{9.10}$$

and

$$F_\beta(-x) \text{ is completely monotone} \quad \Longleftrightarrow \quad \beta \in [\beta_c, \infty).$$

The density is strictly positive at every $x > 0$ whenever $\beta > \beta_c$. At β_c it is nonnegative and has a nonempty finite set of zeros in $(0, \infty)$; all these zeros have finite even order.

Proof. At $\beta = \alpha$, if q_α were nonnegative, then (9.3) and Fatou's lemma would give

$$0 < \int_0^\infty x^{-1} q_\alpha(x)\, dx \leq \liminf_{s \downarrow -1} M_\alpha(s) = 0,$$

because $M_\alpha(s)$ is a positive constant times $(1+s)^{\gamma-1}(1+o(1))$. The left integral is allowed to be infinite, and is strictly positive since the mass is one. Hence complete monotonicity fails at α. By the beta-factor lemma it also fails below α.

We recall why the upper endpoint in (9.10) lies strictly above some positive-density parameters. Put

$$\phi_\beta(t) = \frac{e^{-t}}{1-e^{-t}} - \gamma\frac{e^{-\beta t/\alpha}}{1-e^{-t/\alpha}}.$$

The condition $\phi_\beta \geq 0$ is equivalent to $\beta \geq \beta_{\mathrm{ID}}$. The maximand in (9.1) tends to $-\infty$ at zero and to α from above at infinity; its maximum is attained. It exceeds α. At $\beta_U = \alpha(1+\gamma)/2$,

$$\phi_{\beta_U}(t) > \frac{e^{-t/2}}{2}\left[\frac{1}{\sinh(t/2)} - \frac{\gamma}{\sinh(\gamma t/2)}\right] > 0.$$

The attained maximum is consequently smaller than β_U.

At $\beta = \beta_{\mathrm{ID}}$, the logarithmic gamma identity

$$\log M_\beta(s) = \big(\psi(1) - \alpha\gamma\psi(\beta)\big)s + \int_0^\infty (e^{-st} - 1 + st)\phi_\beta(t)\,\frac{dt}{t}$$

is a Levy–Khintchine representation. Thus $M_\beta(iu)$ is an infinitely divisible characteristic function with exponential Fourier decay. A convolution square root has an analytic nonnegative density on the real line. It is not identically zero, so its real zeros are isolated; its convolution with itself is strictly positive everywhere. Fourier uniqueness then proves $q_\beta(x) > 0$ at all positive x. Lemma 9.1 supplies uniform positive tails in a compact parameter neighborhood. The minimum on the remaining compact x-interval stays positive for all sufficiently nearby parameters. This recovers the strict-gap argument of [116]: some entire interval below β_{ID} is admissible.

The admissible set is an upper set by Lemma 9.2 and is closed in $(0, \infty)$, since every derivative of $F_\beta(-x)$ depends continuously on β. It is therefore $[\beta_c, \infty)$, with $\beta_c > \alpha$; otherwise closedness would make α admissible. The strict gap gives $\beta_c < \beta_{\mathrm{ID}}$.

To obtain strict positivity above the boundary, split the factor in (9.8) as $Z = B^\alpha W$, where $B \sim \mathrm{Beta}(\beta_c, \delta)$, and $W > 0$ is the independent multiplicative convolution power $\gamma - 1$. The density q_{β_c} is analytic and nonnegative, with mass one, so its zeros are isolated. For every $x > 0$,

$$q_{\beta_c+\delta}(x) = \mathbb{E}\big[Z^{-1} q_{\beta_c}(x/Z)\big] > 0.$$

Indeed, conditional on W, the continuous beta variable avoids the discrete zero set with probability one. The integrand is bounded for fixed x by the upper-tail estimate, so the formula is finite.

If q_{β_c} had no zero, uniform positive tails and compactness would preserve positivity below β_c, a contradiction. All its zeros lie in a compact subinterval by Lemma 9.1; analyticity makes their number finite, and nonnegativity makes each order even with positive leading coefficient. □

9.3 The boundary is crossed transversally in the parameter

Theorem 9.4 (Evolution equation and local loss of positivity)**.** *For $\beta > \alpha$,*

$$\partial_\beta q_\beta(x) = \gamma \int_0^\infty \frac{e^{-\beta t}}{1 - e^{-t}} \big[e^{\alpha t} q_\beta(x e^{\alpha t}) - q_\beta(x)\big]\, dt. \tag{9.11}$$

At every zero x_* of q_{β_c},

$$q_{\beta_c}(x_*) = \partial_x q_{\beta_c}(x_*) = 0, \qquad b := \partial_\beta q_{\beta_c}(x_*) > 0.$$

If this zero has order $2m$, and $a = \partial_x^{2m} q_{\beta_c}(x_*)/(2m)! > 0$, then as $\varepsilon \downarrow 0$ a single negative interval is born near x_* for $q_{\beta_c-\varepsilon}$. Its two endpoints satisfy

$$x_\pm(\varepsilon) = x_* \pm (b/a)^{1/(2m)} \varepsilon^{1/(2m)} + o(\varepsilon^{1/(2m)}). \tag{9.12}$$

Its negative mass is

$$\int_{x_-(\varepsilon)}^{x_+(\varepsilon)} -q_{\beta_c-\varepsilon}(x)\, dx = \frac{4m}{2m+1}\, b (b/a)^{1/(2m)} \varepsilon^{1+1/(2m)} (1 + o(1)). \tag{9.13}$$

In particular, an ordinary quadratic contact has a square-root interval width and negative mass of order $\varepsilon^{3/2}$. No assertion that all contacts are quadratic is needed.

Proof. The digamma integral gives

$$\partial_\beta \log M_\beta(s) = \gamma \int_0^\infty \frac{e^{-\beta t}(e^{-\alpha s t} - 1)}{1 - e^{-t}}\, dt.$$

Mellin inversion gives (9.11). Interchanging integrals is justified near $t = 0$ by the cancellation in the brackets and integrability of $(1+|u|)|M_\beta(iu)|$; at infinity the exponential kernel suffices. The resulting real-space integral is also absolutely convergent, by analyticity at small t and the upper tail at large t.

At a zero of the critical density the subtracted term vanishes. The remaining integrand is nonnegative and is positive except at a discrete set of $t > 0$. It has positive integral. Joint analyticity therefore gives, with $h = x - x_*$,

$$q_{\beta_c-\varepsilon}(x_* + h) = ah^{2m} - b\varepsilon + O(h^{2m+1} + \varepsilon|h| + \varepsilon^2).$$

After $h = \varepsilon^{1/(2m)}u$ and division by ε, the local function tends to $au^{2m} - b$, together with its derivatives on compact sets. Analytic root separation, or the Weierstrass preparation theorem, gives exactly two nearby real roots, with the asymptotics (9.12); the other local roots are nonreal. The sign between the real roots is negative. Integration of the scaled limit over $[-(b/a)^{1/(2m)}, (b/a)^{1/(2m)}]$ gives (9.13). □

Corollary 9.5 (The largest-contact characterization)**.** *Define*

$$\mathcal{T} = \{\beta \in (\alpha, \beta_{\mathrm{ID}}) : q_\beta(x) = \partial_x q_\beta(x) = 0 \text{ for some } x > 0\}.$$

This set is nonempty, has a largest element, and

$$\max \mathcal{T} = \beta_c.$$

Consequently complete monotonicity is equivalent to $\beta \geq \max \mathcal{T}$. The original infinite collection of derivative inequalities can thus be replaced by the largest-parameter solution of the explicit two-equation contact system. This is an exact implicit characterization, not an evaluation of its largest solution.

Proof. Theorem 9.3 supplies a contact at β_c. Strict positivity at every larger parameter excludes all larger solutions of the contact system. Smaller contacts, if any, do not affect its largest parameter. □

9.4 An exact hierarchy of finite moment obstructions

For $N \geq 1$, put

$$H_N(\beta) = (m_{i+j}(\beta))_{i,j=0}^N, \qquad K_N(\beta) = (m_{i+j+1}(\beta))_{i,j=0}^N.$$

Theorem 9.6 (Finite thresholds converge, but never reach the boundary)**.** *For each $N \geq 1$ there is a finite $b_N > 0$ such that*

$$H_N(\beta) \succeq 0 \text{ and } K_N(\beta) \succeq 0 \quad \Longleftrightarrow \quad \beta \geq b_N.$$

These thresholds satisfy

$$b_N \leq b_{N+1} < \beta_c, \qquad \lim_{N\to\infty} b_N = \beta_c. \tag{9.14}$$

Thus every non-CM parameter has a finite negative quadratic-form witness in a Hankel or shifted Hankel matrix. Nevertheless every fixed finite collection of these tests accepts some parameters below the true boundary.

Proof. The beta-factor identity makes $H_N(\beta + \delta)$ the entrywise product of $H_N(\beta)$ with the moment matrix of $Z_{\beta,\delta}$. The analogous statement holds for K_N. Schur's product theorem therefore makes the simultaneous positive-semidefinite set an upper set. It is closed, and is nonempty by Theorem 9.3. As $\beta \downarrow 0$, m_1 and m_2 are positive constants times $\beta^{-\gamma}(1 + o(1))$, so $m_2 - m_1^2 < 0$. Hence its left endpoint b_N is positive.

At β_c, both matrices are strictly positive definite: their quadratic forms are respectively $\int p(x)^2 q_{\beta_c}(x)\,dx$ and $\int xp(x)^2 q_{\beta_c}(x)\,dx$, which are positive for every nonzero polynomial p. Continuity of finitely many eigenvalues preserves this property in a parameter neighborhood, proving $b_N < \beta_c$. Taking principal submatrices gives monotonicity in N.

Suppose $b_* := \sup_N b_N < \beta_c$, and choose $b_* < \beta < \beta_c$. Every finite Hankel and shifted Hankel matrix is then positive semidefinite. The classical Stieltjes moment theorem gives a probability measure on $[0, \infty)$ with moments $m_n(\beta)$. For every $t > 0$, monotone convergence gives

$$\mathbb{E}e^{tX} = \sum_{n=0}^{\infty} \frac{m_n(\beta)t^n}{n!} = \Gamma(\beta)^\gamma F_\beta(t) < \infty.$$

Absolute expansion of e^{-tX} therefore identifies its Laplace transform with $\Gamma(\beta)^\gamma F_\beta(-t)$, contradicting the definition of β_c. This proves (9.14). The moment characterization itself is classical and already noted in [3, Section 3]; the strict finite-threshold behavior uses the parameter coupling and the analytic critical density. □

The first unshifted test gives the explicit necessary condition

$$\frac{\Gamma(\beta)\Gamma(\beta + 2\alpha)}{\Gamma(\beta + \alpha)^2} \le 2^{1/\gamma}. \tag{9.15}$$

The left side strictly decreases from infinity to one as β runs from zero to infinity: differentiating its logarithm gives $\psi(\beta) + \psi(\beta + 2\alpha) - 2\psi(\beta + \alpha) < 0$, by strict concavity of the digamma function. Thus this single necessary condition already has a unique, explicit one-dimensional threshold. The shifted 2×2 test similarly requires

$$\frac{\Gamma(\beta + \alpha)\Gamma(\beta + 3\alpha)}{\Gamma(\beta + 2\alpha)^2} \le (3/2)^{1/\gamma}.$$

9.5 A concrete rational obstruction

Theorem 9.7 (Certified lower bound at $\alpha = 1/3, \gamma = 2$)**.** *The actual complete-monotonicity threshold satisfies*

$$\beta_c(1/3, 2) > \frac{428292245841}{10^{12}} = 0.428292245841.$$

Two explicit rational negative-density witnesses are certified. The simpler one is

$$-\frac{1}{2000} < q_{107/250}(1/2) < -\frac{1}{2500} < 0. \tag{9.16}$$

The sharper one, with $\beta_* = 428292245841/10^{12}$ *and* $x_* = 511574841194/10^{12}$*, is*

$$-10^{-12} < q_{\beta_*}(x_*) < -10^{-13} < 0.$$

The bound is certified by a finite interval computation and the explicit analytic remainder estimate below.

Proof. For $\alpha = 1/3$ and $\gamma = 2$, the density has the convergent entire series

$$q_\beta(x) = \Gamma(\beta)^2 \sum_{n=0}^{\infty} \frac{(-x)^n}{n!\,\Gamma(\beta - (n+1)/3)^2}. \tag{9.17}$$

Here a reciprocal Gamma value at a nonpositive integer is interpreted as zero. One justification, avoiding a formal closure of an infinite Mellin contour, is to shift the contour finitely many times to read off $q(0), q'(0), q''(0)$, and then use the Gamma shift relation in the Mellin transform. It gives the ordinary differential equation

$$9q'''(x) + x^2 q''(x) + (9 - 6\beta)xq'(x) + (4 - 3\beta)^2 q(x) = 0.$$

The finite contour shifts give smooth extension at zero and the initial values in (9.17). The equation is regular at zero and has an analytic solution uniquely determined by these three values. Its coefficient

recurrence is exactly the recurrence in (9.17), whose three residue-class series are entire by the ratio test. Uniqueness proves (9.17) globally.

Put $\beta = 107/250$, $x = 1/2$, and

$$t_n = \frac{(1/2)^n}{n!\,\Gamma(107/250 - (n+1)/3)^2} > 0.$$

Gamma's recurrence gives the exact rational ratio

$$\frac{t_{n+3}}{t_n} = \frac{(n+4-321/250)^2}{72(n+1)(n+2)(n+3)} < \frac{n+3}{72(n+1)(n+2)}.$$

For $n \geq 30$, the last expression is at most $33/71424 < 1/1000$. Therefore

$$\left|\sum_{n=30}^{\infty}(-1)^n t_n\right| \leq \frac{1000}{999}(t_{30}+t_{31}+t_{32}) =: T. \tag{9.18}$$

The accompanying script `leroy_rational_certificate.py` uses exact rational inputs and 256-bit Arb balls for the 33 reciprocal Gamma evaluations. It verifies, by interval comparisons,

$$T < 10^{-30}, \qquad \Gamma(107/250)^2\left(\sum_{n=0}^{29}(-1)^n t_n - T\right) > -1/2000,$$

and

$$\Gamma(107/250)^2\left(\sum_{n=0}^{29}(-1)^n t_n + T\right) < -1/2500.$$

For readability, the normalized partial sum is enclosed by the short decimal interval

$$-9.6084728959550860 \times 10^{-5} < \sum_{n=0}^{29}(-1)^n t_n < -9.6084728959550858 \times 10^{-5},$$

and the absolute normalized remainder bound is below 3.838×10^{-31}. The script's rational comparisons, rather than rounded displayed decimals, are the certificate; its log records the full Arb enclosures. Formula (9.18) proves (9.16); the second pair is treated identically. Indeed both pairs satisfy $1/3 < \beta < 1/2$ and $0 < x \leq 13/25$, so for $n \geq 30$ their three-step absolute term ratios are at most

$$(13/25)^3\frac{33}{9\cdot31\cdot32} = \frac{24167}{46500000} < \frac{1}{1000}.$$

For the second pair the absolute normalized tail is below 7.707×10^{-31}; the same script verifies the stated rational interval, enclosing its value near -5.18172×10^{-13}. Finally (9.7) and Theorem 9.3 turn each negative-density witness into the corresponding strict lower bound. No global uniqueness of a numerical contact is assumed. □

9.6 A convex-order bridge to the exponential law

Let X_β have density q_β for $\beta \geq \beta_c$, and set

$$\mu_\beta = \mathbb{E}X_\beta = \frac{\Gamma(\beta)^\gamma}{\Gamma(\beta+\alpha)^\gamma}, \qquad Y_\beta = X_\beta/\mu_\beta.$$

We use $U \leq_{\mathrm{cx}} V$ to mean that U, V have the same mean and $\mathbb{E}\varphi(U) \leq \mathbb{E}\varphi(V)$ for every convex function for which the expectations are defined.

Theorem 9.8 (An exponential upper law in convex order)**.** *If $\beta_c \le \beta_1 < \beta_2$, then*

$$X_{\beta_2} \le_{\mathrm{st}} X_{\beta_1}, \qquad Y_{\beta_1} \le_{\mathrm{cx}} Y_{\beta_2} \le_{\mathrm{cx}} E, \qquad E \sim \mathrm{Exp}(1).$$

Moreover Y_β converges in distribution to E as $\beta \to \infty$. The coupling can be chosen so that $(Y_\beta)_{\beta \ge \beta_c}$ is a positive martingale, bounded in every fixed L^p, and converges in every finite L^p to an exponential terminal variable.

Proof. In (9.9), $0 < Z < 1$, proving stochastic decrease of the unnormalized law. After normalization,

$$Y_{\beta_2} \overset{d}{=} Y_{\beta_1} V, \qquad V = Z/\mathbb{E}Z, \qquad \mathbb{E}V = 1,$$

with independence. Conditional Jensen proves the first convex-order inequality.

For every positive integer n,

$$\mathbb{E}Y_\beta^n = n! \left[\frac{\Gamma(\beta+\alpha)^n}{\Gamma(\beta)^{n-1}\Gamma(\beta+n\alpha)} \right]^\gamma \le n!,$$

where the inequality is Gamma log-convexity. The Gamma-ratio asymptotic makes these moments tend to $n!$. The moment problem for the exponential distribution is determinate; tightness and the uniform higher-moment bounds prove convergence in distribution to E. The second-moment bound supplies uniform integrability, so the expectations of each stop-loss function $(y-t)_+$ converge. Passing to the limit in the convex-order inequalities proves $Y_\beta \le_{\mathrm{cx}} E$.

Finally the logarithms of the factors in (9.8) add when consecutive parameter intervals are concatenated. Independent increments can be realized using a Poisson measure on the parameter and jump variables (b,t) with intensity

$$\gamma \frac{e^{-bt}}{1-e^{-t}}\, db\, dt, \qquad b > \beta_c, \quad t > 0,$$

where a point produces the logarithmic downward jump $-\alpha t$. The integral of $\min(1,t)$ over a finite parameter interval is finite, so this defines the required multiplicative process. Dividing it by its mean gives the conditional-mean-one factors above, and hence a positive martingale Y_β. Uniform bounds in L^n for every integer n give martingale convergence in every fixed finite L^p; its already identified limiting distribution is exponential. □

Corollary 9.9 (Sharp uniform transform bounds)**.** *For every $\beta \ge \beta_c$ and $x > 0$,*

$$e^{-x} < \Gamma(\beta)^\gamma F_\beta(-x/\mu_\beta) < \frac{1}{1+x}.$$

For $0 < x < 1$,

$$e^{x} < \Gamma(\beta)^\gamma F_\beta(x/\mu_\beta) < \frac{1}{1-x}.$$

The upper bounds are sharp in the parameter limit $\beta \to \infty$. Also $\mathbb{E}(Y_\beta - t)_+ \le e^{-t}$ for every $t \ge 0$.

Proof. Apply Jensen and Theorem 9.8 to e^{-xy}, e^{xy}, and $(y-t)_+$. The density is nondegenerate, so Jensen is strict. The independent beta factor is nondegenerate for every positive parameter increment, making each finite-step inequality strict for the exponential test functions; a further parameter increment followed by the limit proves strictness of the upper bounds. For $0 < x < 1$, the uniform moment bound by $n!$ also justifies convergence of the moment-generating series to $(1-x)^{-1}$. □

What remains open here. The results locate a unique true threshold and characterize its finite moment approximations and local density contacts. They do not evaluate β_c in closed form, prove that there is exactly one critical contact, or prove that every contact is quadratic. Numerical solutions of $q = q_x = 0$ are not used to certify the global threshold. The strictly negative rational density value in Theorem 9.7 is a certified lower bound only.

10 Precise open endpoints suggested by the proofs

The following are research targets, not additional proved statements. They are included to show where each bridge reaches and what further argument would actually be needed.

An individual value in the Erdős perturbation class. For an arbitrary integer sequence $a_n/2^{2^n} \to 1$, the common-field theorem proves that at most one rational shift can yield a rational value, and at most two can yield values in a specified quadratic field. Erdős 263 asks for irrationality at the prescribed shift zero throughout that class [7]. Moving from "at most one exception" to "zero is never exceptional" requires an argument that locates the exception. The norm argument does not supply it.

Exact evaluation and geometry of the Le Roy boundary. Theorem 9.3 and Corollary 9.5 reduce the boundary to the largest solution of $q = q_x = 0$. It remains to determine whether the critical density has exactly one zero and whether each critical zero is always quadratic. An affirmative nondegeneracy statement would force the $\varepsilon^{3/2}$ negative-mass law universally. The rigorous lower bound in Theorem 9.7 gives neither an equal upper bound nor a closed expression for β_c.

Effective Bessel selection. The explicit D_r family needs no selection procedure. In contrast, the independent raw K_0 and six-coordinate subsets in Theorem 3.1 are existential. A method that identifies the exceptional arguments, or settles a specified raw $K_0(t)$, would go beyond the rank estimate. No irrationality or transcendence claim for Euler's constant follows from cancelling it.

The binary sign restriction. The prime-power construction gives coefficients in a prescribed congruence class, with exact multiplicity and one minimax solution. Forcing every off-zero coefficient into $\{-1, 1\}$ is an additional nonlinear restriction. The displayed period-81 and period-64 vectors do not meet that restriction, so they are not sign-restricted counterexamples to Erdős-type harmonic problems.

Beyond the nearby-prime binomial family. The four-value histogram depends on $p < q < r$, increasing gaps, and $2(r-p)^2 < p$. Removing those hypotheses, handling arbitrary squarefree integers, or proving a global extremal bound for Erdős 700 requires new information. The exact distribution here gives a family on which such a proposed global result must agree.

From universal to individual integration stability. The geometric iff statements concern all regular functions on an affine curve, or all functions in its algebraic function field. They give a converse to the polynomial-cover construction and a dimension for its first obstruction. They do not classify all individual transcendental elementary functions under repeated integration, the wider setting of [6].

A A scope check using classical Mahler specialization

For completeness, all allowed rational translates of the unperturbed sequence 2^{2^n} are already transcendental by classical Mahler theory. Here is a direct verification of the needed hypotheses; it explains the use of the entire perturbation class in Section 7.

For $h \in \mathbb{Q}$, set

$$g_h(z) = \sum_{n\geq 0} \frac{z^{2^n}}{1 + hz^{2^n}}, \qquad g_h(z) - g_h(z^2) = \frac{z}{1+hz}.$$

This defines an analytic germ at zero and a meromorphic function in the unit disk. It is never rational. Indeed, if $h \neq 0$ and a rational g_h had finite nonzero pole set S, the functional equation would give

$$\{z : z^2 \in S\} \subseteq S \cup \{-1/h\}, \qquad 2|S| \leq |S| + 1.$$

If S is empty, g_h is polynomial, contradicting the pole of the right side. If S has one member, the inclusion forces $S = \{1\}$ and $h = 1$. But a pole of order m at one leaves in the difference a nonzero leading coefficient multiplied by $1 - 2^{-m}$, whereas $z/(1+z)$ is regular at one. This too is impossible. When $h = 0$, the same pole-count argument forces a polynomial, and comparison of degrees in $g(z) - g(z^2) = z$ excludes that possibility.

A nonrational Mahler function is transcendental over $\overline{\mathbb{Q}}(z)$, and Nishioka's specialization theorem transfers the transcendence degree at regular algebraic points [9, Theorem 1]. Apply it to

$$\begin{pmatrix} g_h(z) \\ 1 \end{pmatrix} = \begin{pmatrix} 1 & z/(1+hz) \\ 0 & 1 \end{pmatrix} \begin{pmatrix} g_h(z^2) \\ 1 \end{pmatrix}.$$

The point $1/4$ is regular exactly when $h \neq -2^{2^n}$ for every $n \geq 1$, the original pole exclusions. Thus $g_h(1/4) = \sum_{n\geq 1}(2^{2^n} + h)^{-1}$ is transcendental. For a parameter whose initial power-series disk is small, iterate to a sufficiently small point on this same regular orbit and return by the functional equation; only rational terms are added. This classical corollary is not included in the count of 28 statements.

B Reproducibility and computational certificates

The proof statements concern all allowed inputs. The finite checks listed here audit formulas and examples; they do not replace the infinite-range arguments. The Le Roy sign bound is different in one respect: its numerical part is a rigorous interval certificate, combined with an analytic bound for the infinite remainder.

Check	Result
Corpus identity	130 of 130 Git blob hashes matched the snapshot.
Quadratic weighted digits	4,330 exact prime-valuation cases passed.
Binomial example	All 556,559 indices in the stated half range matched independent Legendre valuations.
Prime 1093	Multiplicative order and both prime-power residues matched.
Period 64 minimum	The algebraic factorization proves height 8 and uniqueness; independent finite enumeration agreed.
Le Roy interval witnesses	Both 256-bit Arb certificates passed; each absolute normalized tail is below 10^{-30}.

The following code is embedded in this single TeX source. The first listing requires Python 3 and `python-flint`; it was executed with `python-flint 0.9.0`. It uses exact rational inputs, reciprocal Gamma intervals, and comparisons against exact rational targets. The second listing requires Python 3 and SymPy. Neither certificate depends on a floating-point root solver.

B.1 Rigorous Le Roy signs

```
"""Rigorous non-CM witness for a Le Roy inverse-Mellin density.

Requires python-flint (tested with 0.9.0 / FLINT 3.6.0).
The mathematical proof of the series and its remainder bound is in
spectral_bridges.tex.  This script certifies only the stated rational
point, and does not locate the global positivity threshold.
"""

import math
```

```python
import flint
from flint import arb, fmpq, ctx


def certify(beta, x, lower_target, upper_target):
    n_terms = 30
    assert fmpq(1, 3) < beta < fmpq(1, 2)
    assert 0 < x <= fmpq(13, 25)

    # All inputs are exact rational numbers, converted to Arb balls.
    terms = [
        arb(x) ** n
        * arb(beta - fmpq(n + 1, 3)).rgamma() ** 2
        / math.factorial(n)
        for n in range(n_terms + 3)
    ]
    partial = sum(
        ((-1) ** n * terms[n] for n in range(n_terms)), arb(0)
    )

    # For n >= 30, beta > 1/3 and x <= 13/25 imply
    # t_(n+3)/t_n <= (13/25)^3 * 33/(9*31*32) < 1/1000.
    assert fmpq(24167, 46500000) < fmpq(1, 1000)
    tail_bound = sum(terms[n_terms : n_terms + 3], arb(0)) * arb(
        fmpq(1000, 999)
    )
    gamma_squared = arb(beta).gamma() ** 2
    lower = gamma_squared * (partial - tail_bound)
    upper = gamma_squared * (partial + tail_bound)

    assert tail_bound < arb(fmpq(1, 10**30))
    assert lower > arb(lower_target)
    assert upper < arb(upper_target)

    print("alpha = 1/3; gamma = 2; beta =", beta, "; x =", x)
    print("Normalized 30-term sum:", partial)
    print("Absolute normalized tail bound:", tail_bound)
    print("q(x) lower enclosure:", lower)
    print("q(x) upper enclosure:", upper)
    print("CERTIFIED:", lower_target, "< q_beta(x) <", upper_target, "< 0")
    print("CONCLUSION: beta_CM(1/3, 2) >", beta)


def main():
    ctx.prec = 256
    print("python-flint version:", flint.__version__)
    print("Arb precision (bits):", ctx.prec)
    certify(fmpq(107, 250), fmpq(1, 2), fmpq(-1, 2000), fmpq(-1, 2500))
    certify(
        fmpq(428292245841, 10**12),
        fmpq(511574841194, 10**12),
        fmpq(-1, 10**12),
        fmpq(-1, 10**13),
    )


if __name__ == "__main__":
    main()
```

The observed certified inequalities are

$$-1/2000 < q_{107/250}(1/2) < -1/2500,$$

and

$$-10^{-12} < q_{428292245841/10^{12}}(511574841194/10^{12}) < -10^{-13}.$$

The normalized absolute remainder bounds were respectively below $3.838 \cdot 10^{-31}$ and $7.707 \cdot 10^{-31}$.

B.2 Exact discrete checks

```
"""Exact supplementary checks for the three discrete/arithmetic bridges.

This is reproducibility material for the accompanying proof dossier.
The all-index theorems have analytic proofs; finite checks do not replace them.
Dependencies: Python 3 and SymPy.
"""

from fractions import Fraction
from itertools import product
from sympy import factorint, n_order


def vp(n, p):
    n = abs(int(n))
    if n == 0:
        raise ValueError("valuation of zero is not needed")
    result = 0
    while n % p == 0:
        n //= p
        result += 1
    return result


def check_quadratic_valuations():
    checked = 0
    for r in range(2, 9):
        modulus = 2**r - 1
        for digits in product((0, 1), repeat=r):
            S = sum(digits[i - 1] * 2**(r - i) for i in range(1, r + 1))
            if S in (0, modulus):
                continue
            y = Fraction(S, modulus)
            denominator = y.denominator
            if int(n_order(2, denominator)) != r:
                continue
            T = sum(i * digits[i - 1] * 2**(r - i) for i in range(1, r + 1))
            U = sum(i*i * digits[i - 1] * 2**(r - i) for i in range(1, r + 1))
            z = Fraction(T, modulus) + Fraction(r*S, modulus**2)
            w = (Fraction(U, modulus) + Fraction(2*r*T, modulus**2)
                 + Fraction(r*r*S*(modulus + 2), modulus**3))
            for prefix_length in range(5):
                value = (prefix_length**2*y + 2*prefix_length*z + w) / 2**prefix_length
                for prime, exponent in factorint(denominator).items():
                    prime, exponent = int(prime), int(exponent)
                    if exponent > vp(r, prime):
                        order = int(n_order(2, prime))
                        lift = vp(2**order - 1, prime)
                        assert vp(value.denominator, prime) == exponent + 2*lift
                        checked += 1
    assert checked == 4330
    return checked


def check_binomial_histogram():
    """Independent of Lucas counting: use factorial valuations at each k."""
    primes = (101, 103, 107)
    n = 101*103*107
    counts = {10403: 0, 10807: 0, 11021: 0, n: 0}
    for k in range(2, n//2 + 1):
        common_divisor = 1
        for prime in primes:
            power, valuation = prime, 0
            while power <= n:
                valuation += n//power - k//power - (n-k)//power
                power *= prime
            if valuation:
                common_divisor *= prime
        assert common_divisor in counts
        counts[common_divisor] += 1
    assert counts == {10403: 1224, 10807: 191, 11021: 116, n: 555028}
    assert sum(counts.values()) == 556559
    return counts


def check_wieferich_example():
```

```
    prime = 1093
    assert all(prime % d for d in range(2, 34))
    assert [pow(2, m, prime) for m in (364, 182, 52, 28)] == [1, 1092, 27, 121]
    assert pow(2, 364, prime**2) == 1
    assert pow(2, 364, prime**3) == 1064432260


if __name__ == "__main__":
    print("Quadratic valuation cases:", check_quadratic_valuations())
    print("Binomial histogram:", check_binomial_histogram())
    check_wieferich_example()
    print("All exact checks passed.")
```

B.3 Compact source manifest

The complete 130-entry source manifest is preserved as comments at the end of this TeX file. The hashes below identify the principal inputs; Git blob SHA-1 is used for repository identity, not as a mathematical validity test.

Manuscript	Git blob SHA-1
[044]	26a390406cc2d22d2cae65b8b67959f8c65b0b9e
[056]	6a45606058e3d5e5349119fe0c03d981b748d011
[080]	b86c948fcde18e637bf7144e627005409a7956ea
[086]	a0c9fb110464b1996c08f1d9d2f11216e98cf862
[088]	9527133f8f4693075e6ec6f08ec19d46bf29eaa9
[112]	625d08cfedfe99dff4810de08f9c632e4bb43172
[113]	8335b4db16c194be3b340f4e06cf4943e5625066
[114]	33f98aaff53b8512ac115ab2202fedeec83f5756
[116]	179f647a09a72ed0c9052b2b1d0dc6f82347abe9

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